Computation and Communication Resource Optimization for Efficient Hierarchical Federated Learning

Chang Liu, *Member, IEEE*, Jun Zhao, *Member, IEEE*, and Yitong Wang

***Abstract*—Federated Learning (FL) is a sophisticated decentralized approach in machine learning, where individual user devices perform model training locally and then transmit their parameters to a central server. This paper delves into the dynamics of a hierarchical FL system, particularly focusing on the intricate exchange of model parameters between cloud and edge servers, and between edge servers and user devices. The capacity of this hierarchical FL model to achieve predefined global model accuracy is significantly influenced by delays in communication and computation of model parameters. To address these challenges and balance the trade-off between energy consumption and execution time, we formulate a comprehensive joint computation and communication resource optimization problem. This problem is aimed at minimizing the weighted sum of FL completion time and total energy consumption, enhancing overall system efficiency. The problem is methodically decomposed into four sub-problems. The initial three sub-problems utilize fractional programming techniques to find efficient solutions that optimize resource allocation. For the final sub-problem, we employ the Concave-Convex Procedure (CCCP), which is well-suited for dealing with the non-linear aspects of the optimization challenge. We then discuss the computational complexity and give the convergence analysis of the proposed approach. Simulation results underline the effectiveness of our approach, showing faster convergence of the global model under optimized communication and computation conditions. These results clearly demonstrate the beneficial trade-off between FL completion time and energy consumption, indicating that our optimization strategy successfully enhances the performance in hierarchical federated learning systems.**

Received 7 June 2024; revised 9 January 2025; accepted 12 August 2025; approved by IEEE TRANSACTIONS ON NETWORKING Editor G. Joshi. This work was supported in part by the National Research Foundation, Singapore, and Infocomm Media Development Authority under its Trust Tech Funding Initiative; in part by the National Research Foundation, Singapore, under its Strategic Capability Research Centres Funding Initiative; in part by the Singapore Ministry of Education Academic Research Fund under Grant Tier 1 RT5/23 and Grant Tier 1 RG90/22; in part by the Nanyang Technological University (NTU)-Wallenberg AI, Autonomous Systems and Software Program (WASP) Joint Project; in part by the Imperial-Nanyang Technological University Collaboration Fund under Grant INCF-2024-008 and Grant INCF-2025-009; in part by the Seatrium New Energy Laboratory Fund (NTU-Seatrium Collaboration) under Grant 03INS002302C130; in part by the National Research Foundation, Prime Minister's Office, Singapore, under its Campus for Research Excellence and Technological Enterprise (CREATE) Program; and in part by the Shanghai Jiao Tong University (SJTU) Startup Fund and the SJTU School of Computer Science. A shorter version of this work containing partial results has been published in the 2022 IEEE/ACM Symposium on Edge Computing (SEC) [1]. *(Corresponding author: Jun Zhao.)*

Chang Liu and Yitong Wang are with the College of Computing and Data Science, Nanyang Technological University, Singapore 639798.

Jun Zhao is with the School of Computer Science, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: Junzhaoemail@ieee.org).

Digital Object Identifier 10.1109/TON.2025.3600163

***Index Terms*—Hierarchical federated learning, edge computing, resource allocation, energy consumption, time minimization.**

I. INTRODUCTION

THE Internet of Things (IoT) and modern mobile devices are producing a lot of data nowadays [2]. These enormous data volumes have fueled the rapid advancement of big data and AI technologies. In order to create a global model, traditional machine learning and deep learning techniques call on devices to transmit their data to a centralized server. However, users are discouraged from uploading data from their local devices to a central server for computation because of dangers concerning privacy-sensitive data leaks. The fast advancement of computing technology has sparked the era of mobile edge computing (MEC), where the processing capacity of mobile device processors is increasing, making it possible to complete increasingly computationally intensive activities like machine learning tasks. Mobile edge devices are replacing servers as the primary computing platform for formerly centralized computing operations. Federated Learning (FL) is a term for decentralized machine learning that takes privacy considerations into account [3], [4]. Through the use of user equipments or user-edge devices, model training is performed locally in FL. The locally learned model parameters are then uploaded to a central server for global model parameter aggregation. The server broadcasts the globally aggregated model to user equipments, and a single FL round is complete. The aforementioned procedure is carried out repeatedly until some stopping criteria is satisfied. The FL method enables devices to create a shared model while protecting user data privacy, since the data never leaves the device.

Federated learning has developed rapidly, yet it still has difficulties with wireless networks. For example, in FL, several participating user devices need to transmit FL parameters across resource-constrained networks, such as wireless networks with constrained bandwidth or power. Therefore, the frequent communication of FL model parameters between user equipment and servers can result in a substantial delay, which reduces the user experience in latency-sensitive applications or in resource-limited systems [5]. Some work has adopted new serverless, decentralized FL models to enhance system robustness, but communication and energy efficiency still need improvement. Specifically, Qu et al. [6] proposed a decentralized federated learning architecture for UAV networks that enhances robustness and flexibility by not relying on a central entity for global model aggregation. Some of these works aim to minimize the overall FL convergence time (delay in both communication and local computation) by allocating resources and solving optimization problems. The focus of papers such as [7] and [8] is to reduce FL convergence time by the optimization of factors like user equipments (UE) uplink bandwidth and device selection. Some other works proposed optimization problems with the aim of reducing FL training loss. In [9], Chen et al. proposed a joint resource allocation and device scheduling optimization problem with the aim of minimizing training loss. Similarly, Wang et al. [10] proposed an optimization problem with the aim of minimizing training loss by a varying number of local update steps between global iterations, and the number of global aggregation rounds. Another genre of papers utilizes the physical properties of wireless communication and proposed solutions such as analog model aggregation over the air (over-the-air computation). These analog aggregation techniques aim to reduce communication latency by allowing devices to upload models simultaneously over a multi-access channel [11], [12], [13]. Yet, in order to aggregate analog models over the air, certain synchronization requirements must be satisfied. Some other methods to reduce communication latency in FL systems include gradient quantization [11], [14] and sparsification [11], other low-rank methods or structured updates [15], [16].

There are also other works of similar genre which proposed optimization problems with the aim of minimizing energy consumption of UE. In [17], the authors proposed an optimization problem which optimizes the local accuracy of edge devices, CPU-cycle frequency of UE, transmit power of UE to base station (BS), and uplink bandwidth of each UE to BS, with the aim to minimize energy consumption of UE. Similarly, the works in [18] aimed to minimize energy consumption of UE by optimizing the uplink transmission time of each user, uplink bandwidth frequency of UE, and device scheduling. However, these works focused on energy consumption and machine learning model performance but failed to consider delay minimization. Reference [19] proposed a joint transmission and computation optimization problem aiming to minimize the total delay in FL, but it only considered the traditional 2-layer FL framework. In [20], although computation latency is taken into account in the objective function, they only optimized the latency and energy for a single round of global aggregation. In contrast, our work optimizes the latency and energy for multiple rounds of global aggregation, thereby enhancing the latency and energy efficiency of the entire federated learning process.

The challenges of minimizing the energy consumption and delay in hierarchical federated learning framework. FL has demanding specifications for device memory, computational power, and communication bandwidth. The devices with limited computational power and limited memory resources will not perform well when training large neural networks. The limited bandwidth also causes clients and servers to experience lengthy delays, which impacts the global model's convergence time. Besides, low-power devices with few computational resources find it challenging to join FL training. Thus, conserving communication and computation resources is critical to increase FL's efficiency. Different from traditional 2-layer FL, in hierarchical FL, the total time is determined not only by UEs, but also by edge servers. To achieve a specific global accuracy, more local iterations provide a more accurate local model while taking more local computation time, but the cost can be mitigated by reducing the number of edge aggregations. More edge aggregations may reduce the demand for local computing, but it incurs more communication delay. Given the above observation, in this paper, we formulate a joint communication and computation optimization problem in order to find the optimal local training iterations, edge aggregation iterations, local CPU frequency, transmission power and communication bandwidth under idealized conditions. While practical deployments face additional complexities such as device dropout, non-IID data, and asynchronous updates, these challenges require specialized strategies that can complement the proposed framework and form an avenue for future research.

Compared with our conference version [1], we have the following improvements:

- **Incorporation of energy consumption into objective function:** In [1], although we have CPU frequency and transmission power in the optimization variables, the energy consumption is not incorporated into the objective function, thus making it always choose the maximum available CPU frequency and transmission power. In this journal version, we extend the optimization problem of [1] by taking energy consumption into consideration in our objective function, and incorporating bandwidth as an optimization variable. In the extended optimization problem, the solutions to CPU frequency and transmission power allow us to gain insight into the trade-off between time and energy in a hierarchical federated learning system.
- **Different problem resolution technique:** This expansion of the problem involves the product of terms containing two optimization variables, which is difficult to resolve. Existing literature typically addresses such issues using the method of alternating optimization, which involves fixing one variable in the product to optimize the other. This approach offers no guarantee of the effectiveness of the solutions obtained. However, in this paper, we use a novel fractional programming technique to tackle this problem, and we are able to find a stationary point for the problem.
- **Improved UE-to-edge association algorithm:** In [1], a greedy-based algorithm was used for the UE-to-edge association problem, which lacked a convergence guarantee. The current paper, however, applies the Concave-Convex Procedure (CCCP) approach to this problem, which is proven to have a convergence guarantee in Section VII.

Contributions. The following list summarizes the major contributions of this paper:

- We formulate a joint communication and computation resource optimization problem that aims to minimize the weighted sum of hierarchical FL energy and time in a hierarchical edge cloud architecture by optimizing the number of local UE computations, the number of edge aggregations, local CPU frequency, transmission power and communication bandwidth under a target global accuracy.
- We consider the edge server model aggregation and edge-to-cloud server transmission delay of multiple rounds of

global aggregation in our optimization objective function. This underscores our comprehensive optimization of the system's entire federated learning process, resulting in notable efficiency improvements.
- We present a UE-to-edge association strategy that aims to minimize the system's latency and energy. The results show that the proposed UE-to-edge association strategy achieves the minimum latency compared with other methods.
- To solve the formulated problem, we divide it into four sub-problems. In tackling the first three sub-problems, we leverage fractional programming techniques for their resolution. For the last sub-problem, we adopt the CCCP method. The simulation results demonstrate that the proposed algorithm can effectively reduce the convergence time in hierarchical federated learning.

The rest of the paper is structured as follows. Section II surveys related work. In Section III, we describe the system model and give a framework for our hierarchical federated learning system. The problem is then formulated in Section IV, followed by the proposed solutions in Section V. We provide a convergence and complexity analysis in Section VI. In Section VII, the numerical results are shown and analyzed. Finally, we conclude the paper in Section VIII.

II. Related Work

There have been a lot of efforts to improve and analyze the performance of federated learning. In this section, we will first present the related work in hierarchical federated learning. After that, we will discuss the works that studied the convergence time of FL and energy consumption in FL.

Hierarchical federated learning. Many studies considered hierarchical federated learning. In [20], Luo et al. introduced a hierarchical FL edge learning framework in contrast to the traditional 2-layer FL systems proposed by the other above-mentioned papers. We should note that there are other papers which propose hierarchical architectures for their FL training that have other aims apart from those mentioned above.

Liu et al. [24] considered a client-edge-cloud hierarchical federated learning system and proposed a novel HierFAVG algorithm which allows edge servers to perform partial model aggregation to enable better communication-computation trade-off while allowing the model to be trained quicker. They also analyzed the 3-layer FL convergence performance under model quantization, and optimized the number of local and edge iterations to reduce the convergence time in [23]. In [25], Wang et al. studied hierarchical federated learning with stochastic gradient descent and conducted a thorough analysis to analyze its convergence behavior. Lim et al. [22] utilized a Stackelberg differential game to model the optimal bandwidth allocation and reward allocation strategies in hierarchical federated learning. Mhaisen et al. [26] utilized branch and bound-based as well as heuristic-based solutions to minimize the data distribution distance at the edge level. For IoT heterogeneous systems, Abdellatif et al. [27] proposed an optimized user assignment and resource allocation solution over hierarchical FL architecture. Huang et al. [28] proposed DAHFL to optimize IoT devices association strategy and allocate computing and communication resources based on dual-distance. It can be seen that hierarchical FL is a promising solution that allows for adaptive and scalable implementation by making use of the resources available at the edge of the network.

Delay minimization in federated learning. The convergence time of FL was studied in many works. Chen et al. [7] jointly considered user selection and resource allocation in cellular networks to reduce the FL convergence time. FedTOE [29] executed a joint allocation of bandwidth and quantization bits to minimize the quantization errors under transmission delay constraint. Sangdeh et al. [30] proposed "CF4FL", a framework that aims to accelerate the convergence speed of the FL training process by designing a deadline-driven vehicle scheduler and a concurrent vehicle polling scheme. Chen et al. [31] proposed to reduce the FL convergence time by reducing the volume of the model parameters exchanged among devices. Yang et al. [19] proposed a joint transmission and computation optimization problem aiming to minimize the total delay in FL. To avoid the training load congestion, Jin et al. [32] formulated a problem to minimize the long-term cumulative latency while guaranteeing the maximal training load and global model convergence. Zhou et al. [33] jointly optimized the energy and delay in FL system by allocating bandwidth, controlling transmission power and CPU frequency of each participating user equipment. At the same time, Wu et al. [34] proposed to minimize the weighted sum of latency and energy by optimizing edge aggregation and association decisions. However, these papers mainly focus on the resource allocation under the constraint of delay or convergence time but they failed to consider how user equipments themselves can reduce computation and communication delay from the frequency of communication between them and the edge servers while maintaining the required machine learning accuracy. Besides, they only studied the traditional 2-layer federated learning framework. In contrast to other papers, Luo et al. [20] utilized a hierarchical model for the optimization problem, which aimed to minimize the weighted combination of delay and UE energy consumption. While they did consider the edge server aggregation and edge-to-cloud transmission delay in their paper, it focused solely on the delay and energy of one single global aggregation round. Conversely, our approach refines the latency and energy across several rounds of global aggregation, leading to improved efficiency in the overall FL convergence procedure.

Energy consumption optimization in federated learning. Some works aim at designing an energy-efficient federated learning. The paper [35] analyzed the probability of correct aggregation in wireless communications and achieved a balance between learning performance and total energy consumption. It assumes that only the server side's Channel State Information (CSI) is available, focusing more on the physical layer's properties. Cui et al. [21] designed a device operating frequency determination approach to optimize energy consumption in a hierarchical FL framework with mobile-edge cloud computing system. Ji et al. [36] proposed an edge device selection and bandwidth allocation approach that minimizes the FL average time cost under energy and delay constraints in the Industrial Internet of Things (IIoT) system. It aims at traditional 2-layer FL in IIoT wireless networks, while our work proposes to minimize the weighted sum of FL time and energy consumption in a hierarchical architecture. Jin et al. [37] designed an online algorithm to control flow selections

TABLE I

COMPARISON BETWEEN THE OPTIMIZATION FRAMEWORK OF THIS PAPER AND RELATED WORKS

	Architecture	**Objective Function**		**Optimization Variables**		
	3-layer FL	Delay Included	Energy Included	The number of local iterations	The number of edge iterations	UE-edge association (using advanced non-greedy approach)
Chen *et al.* [7]	✗	✓	✗	✗	✗	✗
Cui *et al.* [21]	✓	✓	✓	✗	✗	✗
Lim *et al.* [22]	✓	✓	✗	✗	✗	✓
Luo *et al.* [20]	✓	✓	✓	✗	✗	✗
Liu *et al.* [23]	✓	✓	✗	✓	✓	✗
Our SEC version [1]	✓	✓	✗	✓	✓	✗
This paper	✓	✓	✓	✓	✓	✓

and resource provisioning so that the long-term total cost of In-Band Network Telemetry (INT) overhead, resource cost, and federated learning cost are optimized. Wang et al. [38] proposed to maximize the utility of the entire FL cloud-edge system, making the power grid sustainable. Luo et al. [39] developed a low-cost sampling-based algorithm that analyzes and minimizes the total cost while ensuring convergence in FL. However, they focus on optimizing the number of participating clients and the number of local iterations in a 2-layer FL framework without considering controlling the communication parameters such as local CPU frequency, transmission power and bandwidth allocation. The work in [40] considers latency, energy consumption and model accuracy in a NOMA-Enabled hierarchical FL framework in IIoT. However, they fail to consider multiple rounds of global aggregation and take the number of local iterations and edge aggregations into the objective function. The comparison between our work and the works most closely related to it has been summarized in Table I.

Novelty of our work. In this paper, we propose to minimize the weighted sum of FL completion time and energy consumption in a hierarchical FL framework, by optimizing the parameters in both computation and communication. Although the above-mentioned works considered the FL convergence time or energy minimization, they did not take the number of local computations and the number of edge aggregations into account. In our work, under given target accuracy, the proposed method is able to find the optimal number of local computations, the number of edge aggregations, local CPU frequency, transmission power, bandwidth allocation and the UE-to-edge association strategy, thus providing an optimal global setting for efficient hierarchical FL system.

III. SYSTEM MODEL

We consider a hierarchical federated learning model consisting of a cloud server $\mathcal{S}$, a set $\mathcal{M} = \{1, 2, \ldots, M\}$ of M edge servers and a set $\mathcal{N} = \{1, 2, \ldots, N\}$ of N user equipments (UEs) as shown in Fig. 1. Each UE $n \in \mathcal{N}$ owns a local data set $\mathcal{D}_n$ with size D_n.

A. FL Model

In this work, we consider supervised federated learning. $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^{D_n}$ is the training set where $x_i \in \mathbb{R}^d$ is the i-th input sample and $y_i \in \mathbb{R}$ is the corresponding label. Vector

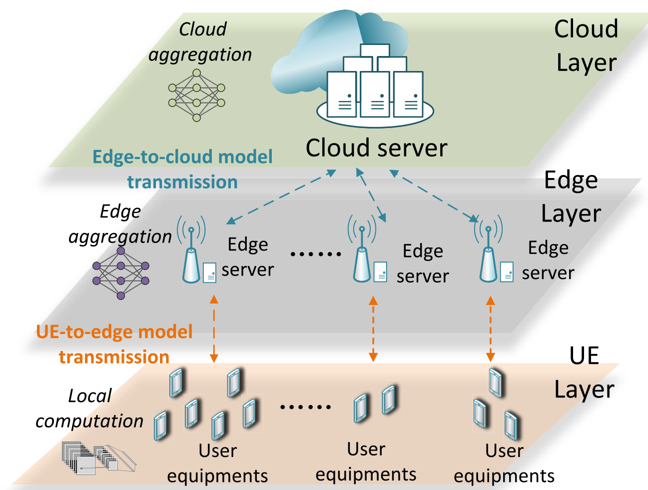


Fig. 1. The architecture of the hierarchical federated learning model.

ω_n is the set of parameters related to the FL model in UE n. We introduce the loss function $f_i(\omega, x_i, y_i)$ that represents the loss function for one sample data. For different learning tasks, the loss function may be different. Let D_n denote the size of the local data set of UE n. The local training is to minimize the loss function $F_n(\omega_n)$ on UE n:

$$\min_{\omega_n} \; F_n(\omega_n) = \frac{\sum_{i \in \mathcal{D}_n} f_i(\omega_n, x_i, y_i)}{D_n}. \tag{1}$$

Let D denote the size of the total data from all the UEs. The training process is to minimize the global loss function $F(\omega)$, which can be formulated as

$$F(\omega) = \frac{\sum_{n=1}^{N} D_n F_n(\omega_n)}{D}. \tag{2}$$

B. Hierarchical Federated Learning Process

The hierarchical federated learning contains five steps: local computation at UE, local model transmission, edge aggregation, edge model transmission and cloud aggregation. Firstly, the UE needs to run several local iterations to achieve a target local accuracy. After that, each UE uploads its local model to the corresponding edge server, and edge server aggregates these models. Then, after a certain number of edge server aggregations, the edge server uploads its model to the cloud,

Algorithm 1 Hierarchical Federated Learning Algorithm

```
Initialize all user equipments (UEs) with parameter ω_n^(0)
i ← 1
while global accuracy ϵ is not obtained do
    for each UE n = 1, 2, ..., N in parallel do
        Update ω_n^(i) = ω_n^(i−1) − η∇F_n(ω_n^(i−1)).
    end for
    if i is an integer multiple of a then
        Each UE l = 1, 2, ..., N in parallel communicate with its corresponding edge server, and edge server performs edge aggregation and broadcasts edge model to the corresponding UEs.
    end if
    if i is an integer multiple of a × b then
        Each edge server m = 1, 2, ..., M in parallel communicate with cloud and cloud performs aggregation and broadcasts it to all the edge servers, and edge server broadcasts edge model to the corresponding UEs.
    end if
    i ← i + 1
end while
```

and the cloud performs global aggregation. The hierarchical FL algorithm is introduced in Algorithm 1. Next, we will give detailed descriptions of the hierarchical FL procedures.

1) Local Computation: Denote $\omega_n^{(i)}$ as the parameters of the local model at i-th local iteration and η as the gradient descent step size. Then the local model is updated as

$$\omega_n^{(i+1)} = \omega_n^{(i)} - \eta \nabla F_n(\omega_n^{(i)}). \tag{3}$$

To measure the time and energy consumption for local training, let f_n be the CPU frequency for the computation of UE n, and C_n be the average number of CPU cycles required for UE n to compute the loss and gradient for one sample data. Following [41], we assume the CPU frequency of UE n remains the same once decided at the start of the local computation process. According to [17], the time required in each iteration for computation of UE n is:

$$t_n^{cmp} = \frac{C_n D_n}{f_n}. \tag{4}$$

Following Lemma 1 in [41], the energy consumption for UE n to run one local iteration is:

$$E_n^{cmp} = \kappa C_n D_n f_n^2, \tag{5}$$

where κ is the effective capacitance coefficient of UE n, which is decided by UE n's chipset architecture. Let a be the number of local iterations for UE to perform in a single round of communication with the corresponding edge server. Then, in one edge round, the total energy UE n consumes is given by aE_n^{cmp}, and the computation time during one round of local training is at_n^{cmp}.

Next, we introduce the subsequent assumption and theorem to discuss the correlation between the local accuracy and the number of local iterations. Adhering to the same Assumption in [17], we present the following assumption.

Assumption 1: $F_n(\omega_n)$ is β-strongly convex and L-smooth. Specifically, $\forall n$ and $\forall \omega_n, \omega_n' \in \mathbb{R}^d$:

$$F_n(\omega_n) \geq F_n(\omega_n') + \left\langle \nabla F_n(\omega_n'), \omega_n - \omega_n' \right\rangle + \frac{\beta}{2} \left\| \omega_n - \omega_n' \right\|^2,$$
$$\left\| \nabla F_n(\omega_n') - \nabla F_n(\omega_n) \right\| \leq L \left\| \omega_n' - \omega_n \right\|,$$

where we consider the ℓ_2 norm of vectors represented by $\|\cdot\|$.

The assumption of strong convexity and smoothness is a widely adopted theoretical simplification in federated learning research. Although neural network loss functions are typically non-convex in practical applications, this assumption facilitates the derivation of convergence guarantees and enables a rigorous analysis of optimization behavior under idealized conditions. These theoretical insights serve as a foundation for understanding the framework's performance and provide critical guidance for the design and optimization of practical FL algorithms.

Theorem 1: Under Assumption 1, in order to achieve a local accuracy[1] $\theta \in (0, 1)$, i.e.,

$$F_n(\omega_n^{(a+1)}) - F_n(\omega_n^*) \leq \theta [F_n(\omega_n^{(0)}) - F_n(\omega_n^*)], \tag{6}$$

where ω_n^* is the optimal parameters of local model that minimizes $F_n(\omega_n)$, with the step size $\eta < \frac{2}{L}$, the number of local iterations a that each UE needs to run is:

$$a = \zeta \ln \frac{1}{\theta}, \tag{7}$$

where $\zeta = \frac{1}{\beta\eta(2 - L\eta)}$.

Proof: The proof is presented in Appendix. □

2) UE-to-Edge Model Transmission: After a local iterations, each UE uploads their local federated learning model to an edge server associated with it. We introduce the indicator variable $\chi_{n,m}$ which represents the association between UE $n \in \mathcal{N}$ and edge server $m \in \mathcal{M}$. Specifically, $\chi_{n,m} = 1$ means that UE n uploads its local federated learning model to edge server m. Otherwise, $\chi_{n,m} = 0$. Each UE can be associated with only one edge server. The user-server association rule can be described as:

$$\begin{cases} \chi_{n,m} \in \{0, 1\}, \forall n \in \mathcal{N}, \forall m \in \mathcal{M}, \\ \sum_{m \in \mathcal{M}} \chi_{n,m} = 1, \forall n \in \mathcal{N}. \end{cases} \tag{8}$$

Without loss of generality, *Frequency Division Multiple Access* (FDMA) communication technique is adopted in this paper. Similar to the setting in [22], we assume each edge server allocates its dedicated bandwidth to the UEs it is associated with. Besides, there is no interference between UEs associated with different edge servers. According to Shannon's formula, the achievable transmission rate of UE n and edge server m can be formulated as

$$r_{n,m} = B_n \log_2 \left(1 + \frac{g_{n,m} p_n}{B_n N_0} \right), \tag{9}$$

where B_n is the bandwidth allocated to UE n, $g_{n,m}$ is the channel gain between UE n and edge server m, p_n is the transmission power of UE n, and N_0 is the noise power. Note that the total bandwidth each edge server m can allocate is $\mathcal{B}_m$, so we have

$$\sum_{n \in \mathcal{N}} \chi_{n,m} B_n \leq \mathcal{B}_m, \forall m \in \mathcal{M}. \tag{10}$$

[1]From Inequality (6), lower θ is better. Yet, the same as prior work [13], [17], [20], this paper still refers to θ as the local accuracy for convenience.

Let d_n denote the size of local model parameters of UE n. In wireless communication, downlink transmission involves broadcast transmission with high server transmission power, resulting in minimal downlink time. In addition, our research does not focus on optimizing server downlink power. Thus, the time for model downloading is omitted. The time cost for transmission local FL model from UE n to its corresponding edge server within one round can be given as follows:

$$t_n^{com} = \sum_{m \in \mathcal{M}} \chi_{n,m} \frac{d_n}{r_{n,m}}. \tag{11}$$

Then, the time for one round of local training (which contains a iterations of local computation) and model transmission for UE n is:

$$T_n^{round} = a t_n^{cmp} + t_n^{com}. \tag{12}$$

Accordingly, the power consumption for the communication between UE n and its corresponding edge server is

$$E_n^{com} = p_n t_n^{com}. \tag{13}$$

3) Edge Aggregation: Let the set of UEs that choose to communicate with edge server m be $\mathcal{N}_m$. When edge server m receives the local models transmitted from its associated UEs, it obtains the averaged parameters ω_m by

$$\omega_m = \frac{\sum_{n \in \mathcal{N}_m} D_n \omega_n}{D_{\mathcal{N}_m}}, \tag{14}$$

where $D_{\mathcal{N}_m} := \sum_{n \in \mathcal{N}_m} D_n$ is the total size of data aggregated at edge server m. Let b be the number of edge aggregations for each edge server to perform in a single round of communication with the cloud. For simplicity, we use "edge iterations" to represent the number of edge aggregations in a single round of communication with the cloud. Let μ denote edge accuracy, i.e., $\tilde{F}(\omega_m^{(b)}) - \tilde{F}(\omega_m^*) \le \mu \left[\tilde{F}(\omega_m^{(0)}) - \tilde{F}(\omega_m^*)\right]$, where $\tilde{F}(\cdot)$ is the loss function on the edge server. To achieve edge accuracy μ, for convex machine learning tasks, the number of edge iterations is given by [42]:

$$b = \frac{\gamma \ln(1/\mu)}{1-\theta}. \tag{15}$$

From (15), it can be observed that b is affected by both edge accuracy μ and local accuracy θ. γ is a constant related to the loss function. When the model is required to be more accurate (i.e., the values of μ and θ are small), the edge needs to run more iterations. After b rounds of edge iterations, the total energy consumption of all the UEs in one cloud round can be calculated as:

$$E^{total} = \sum_{n \in \mathcal{N}} (ab E_n^{cmp} + b E_n^{com}). \tag{16}$$

4) Edge-to-Cloud Model Transmission: After b rounds of edge aggregation, each edge server $m \in \mathcal{M}$ uploads its model parameters ω_m to the cloud and downloads the global model from the cloud again. The delay within one round can be formulated as

$$t_m^{com} = \frac{d_m}{r_m}, \tag{17}$$

where d_m is the size of model parameters at the edge server, and r_m is the transmission rate between edge server m and the cloud.

5) Cloud Aggregation: After every b rounds of edge aggregation, the edge servers send their edge models to the cloud for cloud aggregation. Therefore, the computation and transmission delay during one round of cloud aggregation for edge server m is:

$$T_m^{edge} = b \cdot \max_{n \in \mathcal{N}} \{\chi_{n,m} T_n^{round}\} + t_m^{com}. \tag{18}$$

Finally, the cloud aggregates the model parameters transmitted from the edge servers as follows:

$$\omega = \frac{\sum_{m \in \mathcal{M}} D_{\mathcal{N}_m} \omega_m}{D}, \tag{19}$$

where $D := \sum_{n \in \mathcal{N}} D_n$ is the size of data of all the UEs.

To achieve a certain global accuracy, the number of communications between edge servers and the cloud is determined by both edge accuracy and the target global accuracy. Let the number of communications between edge servers and the cloud be $R(\mu, \epsilon)$, where ϵ is the target global accuracy ϵ. It means that

$$F(\omega_g^{(t)}) - F(\omega_g^*) \le \epsilon \left[F(\omega_g^{(0)}) - F(\omega_g^*)\right], \tag{20}$$

where $\omega_g^{(t)}$ is the parameters of global model at t-th iteration, ω_g^* is the parameters of optimal global model and $F(\cdot)$ is the global loss function.

IV. Optimization for Hierarchical Federated Learning

In this section, we give a formulation of the proposed problem followed by analysis and solutions.

A. Problem Formulation

Given the system model above, we now formulate the hierarchical federated learning communication and computation optimization problem where we aim to optimize the following variables: local iteration a, edge iteration b, local CPU frequency $\boldsymbol{f} := [f_n|_{n \in \mathcal{N}}]$, local transmission power $\boldsymbol{p} := [p_n|_{n \in \mathcal{N}}]$, bandwidth allocation $\boldsymbol{B} := [B_n|_{n \in \mathcal{N}}]$ and UE-to-edge association $\boldsymbol{\chi} := [\chi_{n,m}|_{n \in \mathcal{N}, m \in \mathcal{M}}]$:

$$\min_{a,b,\boldsymbol{f},\boldsymbol{p},\boldsymbol{B},\boldsymbol{\chi}} R(\mu,\epsilon)\Big(\rho_1 \max_{m \in \mathcal{M}} \{T_m^{edge}\} + \rho_2 E^{total}\Big), \tag{21}$$

$$\text{s.t} \quad 0 < p_n \le p_n^{max}, \ \forall n \in \mathcal{N}, \tag{21a}$$

$$\chi_{n,m} \in \{0,1\}, \forall n \in \mathcal{N}, \forall m \in \mathcal{M}, \tag{21b}$$

$$\sum_{m \in \mathcal{M}} \chi_{n,m} = 1, \forall n \in \mathcal{N}, \tag{21c}$$

$$\sum_{n \in \mathcal{N}} \chi_{n,m} B_n \le \mathcal{B}_m, \forall m \in \mathcal{M}, \tag{21d}$$

$$a, b \in \mathbb{N}^+. \tag{21e}$$

Here, $\max_{m \in \mathcal{M}} \{T_m^{edge}\}$ is the time taken for a single round of communication between all edge servers and the cloud. ρ_1 and ρ_2 are weight parameters that indicate the preference on time or energy consumption and $\rho_1 + \rho_2 = 1$. In the objective function, the weighted sum of delay and energy of the entire hierarchical federated learning task is minimized. Constraint (21a) is the maximum transmission power of UEs; constraints (21b) and (21c) are the UE-edge server association rules; constraint (21d) guarantees that the sum of the bandwidths used by UEs communicating to each edge server m does not exceed the bandwidth limit $\mathcal{B}_m$ that edge server m has; constraint (21e) specifies that a and b are positive integers.

First, we introduce Proposition 1 to break down $R(\mu, \epsilon)$.

Proposition 1: The number of communications between edge servers and the cloud, $R(\mu, \epsilon)$, can be determined if the local iterations a and edge iterations b are provided.

Proof: According to (7), given the number of local iterations a, local accuracy θ can be expressed as:

$$\theta = e^{-\frac{a}{\varsigma}}. \tag{22}$$

According to (15), give the number of edge iterations b and ocal accuracy θ, μ can be expressed as:

$$\mu = e^{-\frac{b}{\gamma}(1-\theta)}. \tag{23}$$

Analogous to the relationship between edge servers and UEs, given edge accuracy μ, the number of communications between edge server and cloud $R(\mu, \epsilon)$ can be given by:

$$R(\mu, \epsilon) = \frac{C \ln\left(\frac{1}{\epsilon}\right)}{1-\mu}, \tag{24}$$

where C is a constant related to the loss function. Substituting (22) and (23) into (24), it yields:

$$R(\mu, \epsilon) = \frac{C \ln\left(\frac{1}{\epsilon}\right)}{1 - e^{-\frac{b}{\gamma}\left(1-e^{-\frac{a}{\varsigma}}\right)}}. \tag{25}$$

□

With Proposition 1, we use $R(a, b, \epsilon)$ to represent $R(\mu, \epsilon)$ in the subsequent sections.

Difficulty of solving problem (21): Due to the formulation of t_n^{com} and the integer values of a, b, the optimization problem (21) is non-convex and thus difficult to be exactly solved. In fact, (21) belongs to the so-called "mixed-integer programming" [43], where some decision variables are restricted to be integers while others take continuous values. To tackle (21), further steps are required. First, we introduce two auxiliary variables $\mathcal{T}$ and $\boldsymbol{\tau} = [\tau_m|_{m\in\mathcal{M}}]$, and then problem (21) is equivalent to the following optimization problem:

$$\min_{\mathcal{V}} \ \mathcal{F}(\mathcal{V}) = R(a, b, \epsilon)(\rho_1 \cdot \mathcal{T} + \rho_2 E^{total}), \tag{26}$$

$$\text{s.t. } b\tau_m + t_m^{com} \leq \mathcal{T}, \ \forall m \in \mathcal{M}, \tag{26a}$$

$$\chi_{n,m} T_n^{round} \leq \tau_m, \ \forall n \in \mathcal{N}, \forall m \in \mathcal{M},$$

$$\text{(21a),(21b), (21c),(21d),(21e)} \tag{26b}$$

where $\mathcal{V} := [a, b, \boldsymbol{\tau}, \mathcal{T}, \boldsymbol{f}, \boldsymbol{p}, \boldsymbol{B}, \boldsymbol{\chi}]$. Here, the auxiliary variable $\mathcal{T}$ defines the time interval between each round of communication between edge server m and the cloud, and the auxiliary variable τ_m defines the time interval between each round of communication between UE n and edge server m.

B. Problem Decomposition

In this section, the Block Coordinate Descent (BCD) algorithm will be utilized to solve the original problem. BCD is a technique for optimization that finds a function's minimum by successively minimizing along each block coordinate [44]. First, the problem (26) will be decomposed into four sub-problems. The first sub-problem is the local communication control, which takes transmission power p_n and bandwidth B_n as optimization variables. The second sub-problem is the local computation control, and it has the UE's local iteration number a and UE's CPU frequency f_n as optimization variables. The third sub-problem controls the edge aggregation and obtains the optimal edge iteration number b, slack variable τ_m and $\mathcal{T}$. The last sub-problem optimizes the UE-to-edge association variable $\chi_{n,m}$. The four sub-problems are listed as follows:

Sub-problem I (Local Communication Control):

$$\begin{aligned} &\min\nolimits_{\boldsymbol{p},\boldsymbol{B}} \ \mathcal{F}(\mathcal{V}), \\ &\text{s.t. (26b), (21a), (21d).} \end{aligned} \tag{27}$$

Sub-problem II (Local Computation Control):

$$\begin{aligned} &\min\nolimits_{a,\boldsymbol{f}} \ \mathcal{F}(\mathcal{V}), \\ &\text{s.t. (26b), (21e).} \end{aligned} \tag{28}$$

Sub-problem III (Edge Aggregation Control):

$$\begin{aligned} &\min\nolimits_{b,\boldsymbol{\tau},\mathcal{T}} \ \mathcal{F}(\mathcal{V}), \\ &\text{s.t. (26a), (26b), (21e).} \end{aligned} \tag{29}$$

Sub-problem IV (UE-to-Edge Association):

$$\begin{aligned} &\min\nolimits_{\boldsymbol{\chi}} \ \mathcal{F}(\mathcal{V}), \\ &\text{s.t. (21b), (21c), (21d), (26b).} \end{aligned} \tag{30}$$

V. Optimization Algorithm

In this section, we delve into detailed analysis and propose solutions for the four sub-problems identified in our optimization framework.

A. Optimal Local Communication Control

The formulation of $\sum_{n\in\mathcal{N}} E_n^{com} = \sum_{n\in\mathcal{N}} \frac{p_n d_n}{\sum_{m\in\mathcal{M}} \chi_{n,m} r_{n,m}} = \sum_{n\in\mathcal{N}} \frac{p_n d_n}{\sum_{m\in\mathcal{M}} \chi_{n,m} B_n \log_2\left(1+\frac{g_{n,m} p_n}{B_n N_0}\right)}$ makes Sub-problem II a sum-of-ratios problem which is considered to be NP-hard [45]. To make this part tractable, we add an auxiliary variable $\boldsymbol{\alpha} = [\alpha_n|_{n\in\mathcal{N}}]$ where $\alpha_n \geq \frac{p_n d_n}{\sum_{m\in\mathcal{M}} \chi_{n,m} \cdot r_{n,m}}$. Note that constraint (26b) is equivalent to $r_{n,m} \geq \frac{\sum_{m\in\mathcal{M}} \chi_{n,m} d_n}{\tau_m - a t_n^{cmp}}, \forall n \in \mathcal{N}, m \in \mathcal{M}$. Let $r_{n,m}^{min} = \frac{\sum_{m\in\mathcal{M}} \chi_{n,m} d_n}{\tau_m - a t_n^{cmp}}$. Then, Sub-problem I is equivalent to:

$$\min_{\boldsymbol{p},\boldsymbol{B},\boldsymbol{\alpha}} \sum_{n\in\mathcal{N}} \alpha_n, \tag{31}$$

$$\text{s.t. (21a), (21d).}$$

$$\frac{p_n d_n}{\sum_{m\in\mathcal{M}} \chi_{n,m} \cdot r_{n,m}} \leq \alpha_n, \tag{31a}$$

$$r_{n,m} \geq r_{n,m}^{min}, \forall n \in \mathcal{N}, m \in \mathcal{M}. \tag{31b}$$

Theorem 2: If $(p_n^*, B_n^*, \alpha_n^*)$ is the solution to problem (31), then there exists ψ_n^* satisfying that (p_n^*, B_n^*) is a solution to the following problem with $\psi_n = \psi_n^*$ and $\alpha_n = \alpha_n^*$:

$$\begin{aligned} &\min_{\boldsymbol{p},\boldsymbol{B}} \sum_{n\in\mathcal{N}} \psi_n \left(p_n d_n - \alpha_n \sum_{m\in\mathcal{M}} \chi_{n,m} r_{n,m} \right), \\ &\text{s.t. (21a), (21d), (31b).} \end{aligned} \tag{32}$$

Moreover, when $\psi_n = \psi_n^*$ and $\alpha_n = \alpha_n^*$, p_n^* satisfies $\psi_n^* = \frac{1}{\sum_{m\in\mathcal{M}} \chi_{n,m} \cdot r_{n,m}}$ and $\alpha_n = \frac{p_n^* d_n}{\sum_{m\in\mathcal{M}} \chi_{n,m} \cdot r_{n,m}}$.

Proof: First, we show that $\sum_{m\in\mathcal{M}}\chi_{n,m}r_{n,m}$ is concave with respect to (w.r.t.) p_n and B_n. Let $\ell(p_n,B_n)=\sum_{m\in\mathcal{M}}\chi_{n,m}r_{n,m}$. Then, the Hessian matrix is:

$$Hessian(\ell)=\begin{bmatrix} -\frac{\sum_{m\in\mathcal{M}}\chi_{n,m}g_{n,m}^2}{\ln 2\cdot B_n N_0^2\left(1+\frac{g_{n,m}p_n}{B_n N_0}\right)^2} & \frac{\sum_{m\in\mathcal{M}}\chi_{n,m}g_{n,m}^2 p_n}{\ln 2\cdot(B_n N_0+g_{n,m}p_n)^2} \\ \frac{\sum_{m\in\mathcal{M}}\chi_{n,m}g_{n,m}^2 p_n}{\ln 2\cdot(B_n N_0+g_{n,m}p_n)^2} & -\frac{\sum_{m\in\mathcal{M}}\chi_{n,m}g_{n,m}^2 p_n^2}{\ln 2\cdot B_n(B_n N_0+g_{n,m}p_n)^2}\end{bmatrix}. \quad (33)$$

Given any vector $\boldsymbol{x}=[x_1,x_2]^\intercal\in\mathbb{R}^2$, it can be observed that:

$$\boldsymbol{x}^\intercal Hessian(\ell)\boldsymbol{x}=-\frac{\sum_{m\in\mathcal{M}}\chi_{n,m}g_{n,m}^2(B_n x_1-p_n x_2)^2}{\ln 2\cdot B_n(B_n N_0+g_{n,m}p_n)^2}\le 0.$$

Therefore, $Hessian(\ell)$ is a negative semidefinite matrix. Thus, $\sum_{m\in\mathcal{M}}\chi_{n,m}r_{n,m}$ is concave w.r.t. p_n and B_n. With $\sum_{m\in\mathcal{M}}\chi_{n,m}r_{n,m}$ being concave, the remaining detailed proof is provided in Lemma 2.1 in [45]. □

According to Theorem 2, it is evident that Sub-problem I possesses an optimal solution which coincides with that of problem (32). Since $\sum_{m\in\mathcal{M}}\chi_{n,m}r_{n,m}$ is concave w.r.t. p_n and B_n, the objective function associated with problem (32) is convex. The constraints (21a), (21d), (31b) are all convex. Then, problem (32) belongs to convex optimization and hence we consider the Karush-Kuhn-Tucker (KKT) conditions as follows to obtain the structural features of the optimal solution. First, with $\boldsymbol{\upsilon}:=[\upsilon_{1,1},\ldots,\upsilon_{N,M}]$, $\boldsymbol{\xi}:=[\xi_1,\ldots,\xi_N]$ and $\boldsymbol{\nu}=[\nu_1,\ldots,\nu_M]$ denoting the Lagrange multipliers, the Lagrangian function is given as:

$$\begin{aligned}\mathcal{L}(\boldsymbol{p},\boldsymbol{B},\boldsymbol{\upsilon},\boldsymbol{\xi},\boldsymbol{\nu})=&\sum_{n\in\mathcal{N}}\psi_n\left(p_n d_n-\alpha_n\sum_{m\in\mathcal{M}}\chi_{n,m}r_{n,m}\right)-\\ &\sum_{n\in\mathcal{N},m\in\mathcal{M}}\upsilon_{n,m}\left(B_n\log_2\left(1+\frac{g_{n,m}p_n}{B_n N_0}\right)-r_{n,m}^{min}\right)+\\ &\sum_{n\in\mathcal{N}}\xi_n(p_n-p_n^{max})+\sum_{m\in\mathcal{M}}\nu_m\left(\sum_{n\in\mathcal{N}}\chi_{n,m}B_n-\mathcal{B}_m\right).\end{aligned} \quad (34)$$

After applying KKT conditions, we can get:

$$\begin{aligned}\frac{\partial L}{\partial p_n}=&\psi_n\left(d_n-\alpha_n\sum_{m\in\mathcal{M}}\chi_{n,m}\frac{g_{n,m}}{N_0\ln 2\cdot\left(1+\frac{g_{n,m}p_n}{B_n N_0}\right)}\right)-\\ &\sum_{m\in\mathcal{M}}\frac{\upsilon_{n,m}g_{n,m}}{N_0\ln 2\cdot\left(1+\frac{g_{n,m}p_n}{B_n N_0}\right)}+\xi_n=0,\end{aligned} \quad (35)$$

$$\begin{aligned}\frac{\partial L}{\partial B_n}=&\left(\psi_n\alpha_n\sum_{m\in\mathcal{M}}\chi_{n,m}+\upsilon_{n,m}\right)\cdot\\ &\left(\frac{g_{n,m}p_n}{\ln 2\cdot\left(1+\frac{g_{n,m}p_n}{B_n N_0}\right)B_n N_0}-\log_2\left(1+\frac{g_{n,m}p_n}{B_n N_0}\right)\right)+\nu=0,\end{aligned} \quad (36)$$

$$-\upsilon_{n,m}\left(B_n\log_2\left(1+\frac{g_{n,m}p_n}{B_n N_0}\right)-r_{n,m}^{min}\right)=0, \quad (37)$$

$$\nu_m\left(\sum_{n\in\mathcal{N}}\chi_{n,m}B_n-\mathcal{B}_m\right)=0. \quad (38)$$

Then, p_n and B_n are given by:

$$p_n=\frac{\sum_{m\in\mathcal{M}}(\upsilon_{n,m}+\psi_n\alpha_n\chi_{n,m})B_n}{(\psi_n d_n+\xi)\ln 2}-\frac{B_n N_0}{g_{n,m}}, \quad (39)$$

$$B_n=\sum_{m\in\mathcal{M}}\frac{\chi_{n,m}d_n}{\log_2\left(\frac{(\psi_n\alpha_n+\upsilon_{n,m})g_{n,m}}{\ln_2 d_n\psi_n N_0}\right)(\tau_m-at_n^{cmp})}. \quad (40)$$

Lagrange multipliers update. The optimal solution to the bandwidth allocation B_n and transmission power p_n can be calculated through (39) and (40) if the Lagrange dual variables are given. The dual function of problem (32) is given by $g(\boldsymbol{\upsilon},\boldsymbol{\xi},\boldsymbol{\nu})=\inf_{\boldsymbol{p},\boldsymbol{B}}\mathcal{L}(\boldsymbol{p},\boldsymbol{B},\boldsymbol{\upsilon},\boldsymbol{\xi},\boldsymbol{\nu})$. The Lagrange dual variables $\boldsymbol{\upsilon}$, $\boldsymbol{\xi}$ and $\boldsymbol{\nu}$ can be obtained by solving the Lagrange dual problem of problem (32), which can be expressed as follows:

$$\max_{\boldsymbol{\upsilon},\boldsymbol{\xi},\boldsymbol{\nu}}\ g(\boldsymbol{\upsilon},\boldsymbol{\xi},\boldsymbol{\nu}), \quad (41)$$

$$\text{s.t. } \upsilon_{n,m},\xi_n,\nu_m\ge 0,\forall n\in\mathcal{N},m\in\mathcal{M}. \quad (41a)$$

The Lagrange dual problem is a convex problem, which can be solved by subgradient projection method [46]. The gradients of $g(\boldsymbol{\upsilon},\boldsymbol{\xi},\boldsymbol{\nu})$ can be given by:

$$\begin{aligned}\nabla\upsilon_{n,m}&=B_n\log_2\left(1+\frac{g_{n,m}p_n}{B_n N_0}\right)-r_{n,m}^{min},\\ \nabla\nu_m&=\sum_{n\in\mathcal{N}}\chi_{n,m}B_n-\mathcal{B}_m,\\ \nabla\xi_n&=p_n-p_n^{max}.\end{aligned} \quad (42)$$

Having obtained the subgradients of σ_m, the Lagrange multipliers are updated using the subgradient projection method with a specified step size ϕ. This update process is articulated as follows:

$$\begin{aligned}\upsilon_{n,m}^{(i+1)}&=\max\{\upsilon_{n,m}^{(i)}+\phi\nabla\upsilon_{n,m}^{(i)},0\},\\ \nu_m^{(i+1)}&=\max\{\nu_m^{(i)}+\phi\nabla\nu_m^{(i)},0\},\\ \xi_n^{(i+1)}&=\max\{\xi_n^{(i)}+\phi\nabla\xi_n^{(i)},0\},\end{aligned} \quad (43)$$

where i denotes the number of iterations. This formula ensures that the dual variables, $\upsilon_{n,m}$, ξ_n and ν_m, are non-negative at each iteration, conforming to the constraints imposed by the optimization problem. The iterative update of these Lagrangian dual variables via the subgradient projection method is not only a systematic approach but also ensures convergence under appropriate conditions, such as a diminishing step size ϕ. Each iteration of the algorithm refines the estimates of the dual variables, gradually moving towards the optimal values that satisfy the primal-dual optimality conditions. The process is summarized in Algorithm 2.

B. Optimal Local Computation Control

In this section, we design an algorithm to solve the min-max Sub-problem II. Sub-problem II falls into the category of integer programming which is NP-hard problem [47] since a is a positive integer variable. While integer programming is considered to be an NP-hard problem, we explore the approach of relaxing the integer constraint. This involves treating the variable a as continuous initially and later rounding it back to an integer. Besides, the specific constraint (26b) is equivalent to $f_n\ge\chi_{n,m}\frac{aC_nD_n}{\tau_m-t_n^{com}},\forall n\in\mathcal{N},m\in\mathcal{M}$. Consequently, this

Algorithm 2 Bandwidth Allocation Optimization and Transmission Power Control

Initialize feasible Lagrange dual variables $\boldsymbol{v}^{(0)}, \boldsymbol{\xi}^{(0)}, \boldsymbol{\nu}^{(0)}$, the target accuracy ϵ_1 and maximum number of iteration i_1.
$i \leftarrow 1$
while $i \leq i_1$ **do**
  Update $\boldsymbol{B}^{(i)}$ and $\boldsymbol{p}^{(i)}$ according to (39) and (40) using $\boldsymbol{v}^{(i)}, \boldsymbol{\xi}^{(i)}, \boldsymbol{\nu}^{(i)}$.
  Update Lagrangian dual variables $\boldsymbol{v}^{(i+1)}, \boldsymbol{\xi}^{(i+1)}, \boldsymbol{\nu}^{(i+1)}$ using $\boldsymbol{B}^{(i)}$ and $\boldsymbol{p}^{(i)}$ according to (43).
  if $\left\|\boldsymbol{v}^{(i+1)} - \boldsymbol{v}^{(i)}\right\| < \epsilon_1, \left\|\boldsymbol{\xi}^{(i+1)} - \boldsymbol{\xi}^{(i)}\right\| < \epsilon_1, \left\|\boldsymbol{\nu}^{(i+1)} - \boldsymbol{\nu}^{(i)}\right\| < \epsilon_1$ **then**
    $\boldsymbol{B}^* = \boldsymbol{B}^{(i)}$ and $\boldsymbol{p}^* = \boldsymbol{p}^{(i)}$.
    Break.
  else
    $i \leftarrow i+1$
  end if
end while
Output: $\boldsymbol{B}^*$ and $\boldsymbol{p}^*$.

leads to a revised perspective on the lower bound of energy consumption for computation as follows:

$$E_n^{cmp} \geq \kappa C_n D_n \left(\chi_{n,m} \frac{a C_n D_n}{\tau_m - t_n^{com}}\right)^2, \forall n \in \mathcal{N}, m \in \mathcal{M}. \tag{44}$$

By substituting (44) into the original formulation of Sub-problem II in (28), we achieve a reformulated version of Sub-problem II:

$$\min_{a,\boldsymbol{f}} \ R(a,b,\epsilon)\Bigg(\rho_1 \mathcal{T} + \rho_2 ab \sum_{m\in\mathcal{M}} \sum_{n\in\mathcal{N}} \kappa C_n D_n \left(\chi_{n,m} \frac{a C_n D_n}{\tau_m - t_n^{com}}\right)^2 + \rho_2 b \sum_{n\in\mathcal{N}} E_n^{com}\Bigg), \tag{45}$$

$$\text{s.t. } a \geq 1. \tag{45a}$$

Theorem 3: We rewrite the objective function of problem (45) as $\min_a \frac{A(a)}{B(a)}$ where $A(a) = C\ln\left(\frac{1}{\epsilon}\right) \left(\rho_1 \mathcal{T} + \rho_2 ab \sum_{m\in\mathcal{M}} \sum_{n\in\mathcal{N}} \kappa C_n D_n \left(\chi_{n,m} \frac{a C_n D_n}{\tau_m - t_n^{com}}\right)^2 + \rho_2 b \sum_{n\in\mathcal{N}} E_n^{com}\right)$ and $B(a) = 1 - e^{-\frac{b}{\gamma}\left(1 - e^{-\frac{a}{\zeta}}\right)}$. Then, we can obtain a stationary point a^* for problem (45) if alternating optimizing a and y_1 of the following optimization problem converges to (a^*, y_1^*):

$$\begin{aligned} \min_{a,y_1} \ & [A(a)]^2 y_1 + \frac{1}{4[B(a)]^2 y_1}, \\ \text{s.t. } & (45a), \\ & y_1 \in \mathbb{R}^+. \end{aligned} \tag{46}$$

Proof: First, we show that $B(a)$ is concave w.r.t a. Let $g(x) = 1 - e^{-x}$, and then $B(a)$ can be expressed as

$$B(a) = g\left[\frac{b}{\gamma} \cdot g\left(\frac{a}{\zeta}\right)\right]$$

.The second-order partial derivatives $\frac{\mathrm{d}^2 B}{\mathrm{d}a^2}$ can be given by:

$$\frac{\mathrm{d}^2 B}{\mathrm{d}a^2} = \frac{b}{\gamma\zeta}\left\{\frac{b}{\gamma\zeta} g''\left[\frac{b}{\gamma} g\left(\frac{a}{\zeta}\right)\right] g'^2\left(\frac{a}{\zeta}\right) + \frac{1}{\zeta} g'\left[\frac{b}{\gamma} g\left(\frac{a}{\zeta}\right)\right] g''\left(\frac{a}{\zeta}\right)\right\}. \tag{47}$$

Note that $g(x) = 1 - e^{-x}$, $g'(x) = e^{-x}$ and $g''(x) = -e^{-x}$. Thus, $g''(x) = -g'(x)$. Then, $\frac{\mathrm{d}^2 B}{\mathrm{d}a^2}$ can be rewritten as:

$$\frac{\mathrm{d}^2 B}{\mathrm{d}a^2} = \frac{b}{\gamma\zeta^2} g'\left(\frac{a}{\zeta}\right) g'\left[\frac{b}{\gamma} g\left(\frac{a}{\zeta}\right)\right]\left[-\frac{b}{\gamma} g'\left(\frac{a}{\zeta}\right) - 1\right]. \tag{48}$$

From $g(x) = 1 - e^{-x}$ and $g'(x) = e^{-x}$, we have for any x, $g(x) > 0$ and $g'(x) > 0$. $b > 0, \gamma > 0$, so $-\frac{b}{\gamma} g'\left(\frac{a}{\zeta}\right) - 1 < 0$. Therefore, $\frac{\mathrm{d}^2 B}{\mathrm{d}a^2} < 0$ and $B(a)$ is concave w.r.t. a.

Besides, it is easy to verify that $A(a)$ is convex w.r.t. a when $a \geq 0$. With convex $A(a)$ and concave $B(a)$, the remaining proof can be found in Appendix of [48]. □

Since $B(a) > 0$, one can easily verify that $\frac{1}{4[B(a)]^2 y_1}$ is convex w.r.t. a. Therefore, if y_1 is given, problem (46) is convex. Thus, the same Lagrange multipliers procedure as in Section V-A can be applied to solve this problem. Next, we use alternating optimization (AO) to optimize a and y_1 by solving problem (46) iteratively. AO means looping through these two steps until convergence; i.e., the difference in the objective function values between successive iterations is within a predefined error tolerance. Specifically, we first initialize a feasible $a^{(0)}$ as a starting point. Then, $y_1^{(0)}$ is set as $A(a^{(0)})B(a^{(0)})$ (i.e., optimizing y_1 by solving problem (46) with fixed a). We then optimize a by solving problem (46) and get $a^{(1)}$. In i-th iteration, $y_1^{(i)}$ is set as $A(a^{(i)})B(a^{(i)})$, and $a^{(i+1)}$ is obtained by solving problem (46) with fixed $y_1^{(i)}$. The AO is proved to converge according to Section V-C in [48]. After obtaining a, we derive f_n from constraint (26b):

$$f_n = \max_{m\in\mathcal{M}} \left\{\chi_{n,m} \frac{a C_n D_n}{\tau_m - t_n^{com}}\right\}, \forall n \in \mathcal{N}. \tag{49}$$

C. Optimal Edge Aggregation Control

In this problem, $\mathcal{T}$ defines the time interval between each round of communication between edge server m and the cloud, and τ_m defines the time interval between each round of communication between UE n and edge server m. Thus, once a and b are given, τ_m and $\mathcal{T}$ can be obtained from constraints (26a) and (26b). To this end, we first obtain b by solving Sub-problem III, and then derive τ_m and $\mathcal{T}$. Firstly, we relax the integer constraint by allowing b to be a continuous variable, which will be rounded back to an integer number later.

Theorem 4: We rewrite the objective function of Sub-problem III as $min_b \frac{C(b)}{D(b)}$ where $C(b) = C\ln\left(\frac{1}{\epsilon}\right)\left(\rho_1 \mathcal{T} + \rho_2\left(ab \sum_{n\in\mathcal{N}} E_n^{cmp} + b \sum_{n\in\mathcal{N}} E_n^{com}\right)\right)$ and $D(b) = 1 - e^{-\frac{b}{\gamma}\left(1 - e^{-\frac{a}{\zeta}}\right)}$. Then, we can obtain a stationary point for Sub-problem III if the following optimization problem converges to (b^*, y_2^*):

$$\begin{aligned} \min_{b,y_2} \ & [C(b)]^2 y_2 + \frac{1}{4[D(b)]^2 y_2}, \\ \text{s.t. } & b \geq 1, \\ & y_2 \in \mathbb{R}^+. \end{aligned} \tag{50}$$

Proof: First, we show that $D(b)$ is concave w.r.t b. Let $g(x) = 1 - e^{-x}$, and then $D(b)$ can be expressed as $D(b) = g\left[\frac{b}{\gamma} \cdot g\left(\frac{a}{\zeta}\right)\right]$. The concavity of $D(b)$ can be obtained by calculating the second-order partial derivatives:

$$\frac{\mathrm{d}^2 D}{\mathrm{d}b^2} = \frac{1}{\gamma^2} g'' \left[\frac{b}{\gamma} g\left(\frac{a}{\zeta}\right)\right] g^2\left(\frac{a}{\zeta}\right). \tag{51}$$

Since $g(x) = 1 - e^{-x}$, $g'(x) = e^{-x}$ and $g''(x) = -e^{-x}$. Thus, $g''(x) = -g'(x)$, $\frac{\mathrm{d}^2 D}{\mathrm{d}b^2}$ is rewritten as:

$$\frac{\mathrm{d}^2 D}{\mathrm{d}b^2} = -\left[\frac{1}{\gamma} g\left(\frac{a}{\zeta}\right)\right]^2 g'\left[\frac{b}{\gamma} g\left(\frac{a}{\zeta}\right)\right]$$

.

It is obvious that $\frac{\mathrm{d}^2 D}{\mathrm{d}b^2} < 0$. Thus, $D(b)$ is concave w.r.t b. Besides, one can readily confirm that $C(b)$ is convex w.r.t b when $b \geq 0$. With convex $C(b)$ and concave $D(b)$, the remaining proof can be found in Appendix of [48]. □

We use the same AO approach to optimize y_2 and b iteratively. In i-th iteration, $y_2^{(i)}$ is set as $C(b^{(i)}) \cdot D(b^{(i)})$ (i.e., optimizing y_2 by solving problem (50) with fixed b), and $b^{(i+1)}$ is obtained by solving problem (50) with fixed $y_2^{(i)}$. The AO is proved to converge according to Section V-C in [48].

Solution to $\mathcal{T}$ and $\boldsymbol{\tau}$. The optimal solution of a^*, b^* have been obtained with the specified accuracy ϵ. Building upon these results, we can now determine the optimal solution to $\boldsymbol{\tau}$ and $\mathcal{T}$ in accordance with constraints (26a) and (26b). The optimal solution for $\boldsymbol{\tau}$ is derived as follows:

$$\tau_m^* = \max_{n\in\mathcal{N}}\{\chi_{n,m}(a^* \cdot t_n^{cmp} + t_n^{com})\}. \tag{52}$$

Subsequently, the overall system latency $\mathcal{T}$ is computed by considering the maximum of the adjusted latencies across all edge servers, which incorporates the number of edge iterations b and the communication time from the UEs to the edge server:

$$\mathcal{T}^* = \max_{m\in\mathcal{M}}\{b^* \cdot \tau_m^* + t_m^{com}\}. \tag{53}$$

D. Optimal UE-to-Edge Association

In this part, we present the optimal UE-to-edge association scheme. With the optimal a, b, f_n, p_n, B_n, $\boldsymbol{\tau}$ and $\mathcal{T}$, the UE-to-edge association problem can be simplified to the following problem:

$$\min_{\boldsymbol{\chi}} \rho_1 \mathcal{T} + \sum_{n\in\mathcal{N}} \rho_2 b E_n^{com}$$
$$\text{s.t. (21b), (21c), (21d), (26b).} \tag{54}$$

Problem (54) is now a *mixed integer linear programming* (MILP) problem with the objective of minimizing the weighted sum of min-max and min-sum. To solve the UE-to-edge association problem practically, we transform the non-convex function in the objective as convex constraints by introducing a new variable z. Without loss of equivalence, problem (54) can be written as:

$$\min_{z,\boldsymbol{\chi}} \ z, \tag{55}$$
$$\text{s.t. } z \geq \rho_1\mathcal{T} + \sum_{n\in\mathcal{N}} \rho_2 b E_n^{com}, \tag{55a}$$
$$0 \leq \chi_{n,m} \leq 1, \forall n \in \mathcal{N}, m \in \mathcal{M}, \tag{55b}$$
$$\sum_{n\in\mathcal{N}}\sum_{m\in\mathcal{M}} \chi_{n,m}(1-\chi_{n,m}) \leq 0,$$
$$\text{(21c), (21d), (26b).} \tag{55c}$$

Note that constraint (21b) is equivalent to the combination of constraints (55b) and (55c). However, constraint (55c) is not convex. Therefore, it is still challenging to solve problem (55) efficiently. Next, we penalize the concave constraint (55c) to the objective function with penalty parameter $\varphi > 0$:

$$\min_{\boldsymbol{\chi}} \ z - \varphi \sum_{n\in\mathcal{N}}\sum_{m\in\mathcal{M}} \chi_{n,m}(\chi_{n,m}-1),$$
$$\text{s.t. (55a),(55b),(21c),(21d), (26b).} \tag{56}$$

Problem (56) can be handled with CCCP technique by approximating $\sum_{n\in\mathcal{N}}\sum_{m\in\mathcal{M}} \chi_{n,m}(\chi_{n,m}-1)$ with the first-order Taylor series approximation [49]:

$$\sum_{n\in\mathcal{N}}\sum_{m\in\mathcal{M}}\chi_{n,m}^{(i)}(\chi_{n,m}^{(i)}-1)+\sum_{n\in\mathcal{N}}\sum_{m\in\mathcal{M}}(2\chi_{n,m}^{(i)}-1)(\chi_{n,m}-\chi_{n,m}^{(i)}), \tag{57}$$

where $\chi_{n,m}^{(i)}$ represents the approximation of $\chi_{n,m}$ at the i-th iteration. To solve the optimization problem in (55), it requires to initialize the UE-to-edge association $\chi_{n,m}^{(0)}$. Then, the CCCP method is utilized to solve it iteratively until a pre-defined threshold is met. Denote $\widetilde{F}(\varphi)$ as the optimal value of the objective function of problem (56) with penalty parameter φ. The penalty parameter φ is chosen to be a sufficiently large value such that

$$\varphi \geq \frac{z^0 - \widetilde{F}(0)}{\max_{\chi_{n,m}}\left\{\sum_{n\in\mathcal{N}}\sum_{m\in\mathcal{M}}\chi_{n,m}(\chi_{n,m}-1)\right\}}, \tag{58}$$

where z^0 is calculated with any $\chi_{n,m}^{(0)}$ satisfying constraint (55b). Then, problem (55) is equivalent to problem (56) according to Theorem 1 in [50]. Note that we may not always obtain a feasible solution to problem (55). The one that achieves the minimum objective value after performing multiple CCCP with different initial feasible points of problem (56) will be chosen as the final solution.

Let $\mathcal{F}$ denote the objective function in problem (26). The complete communication and computation design algorithm based on BCD is presented in Algorithm 3.

VI. Convergence and Time Complexity

In this section, we analyze the convergence of the proposed algorithm and present the time complexity.

Convergence. We first show Algorithm 3 makes progress in minimizing $\mathcal{F}(\mathcal{V}_1, \mathcal{V}_2, \mathcal{V}_3, \mathcal{V}_4)$ in each iteration; i.e., $\mathcal{F}(\mathcal{V}_1^{(t)}, \mathcal{V}_2^{(t)}, \mathcal{V}_3^{(t)}, \mathcal{V}_4^{(t)})$ is non-increasing w.r.t. t. Specifically, we obtain

$$\begin{aligned}
&\mathcal{F}(\mathcal{V}_1^{(t-1)}, \mathcal{V}_2^{(t-1)}, \mathcal{V}_3^{(t-1)}, \mathcal{V}_4^{(t-1)}) \\
&\overset{\text{(a)}}{\geq} \mathcal{F}(\mathcal{V}_1^{(t)}, \mathcal{V}_2^{(t-1)}, \mathcal{V}_3^{(t-1)}, \mathcal{V}_4^{(t-1)}) \\
&\overset{\text{(b)}}{\geq} \mathcal{F}(\mathcal{V}_1^{(t)}, \mathcal{V}_2^{(t)}, \mathcal{V}_3^{(t-1)}, \mathcal{V}_4^{(t-1)}) \\
&\overset{\text{(c)}}{\geq} \mathcal{F}(\mathcal{V}_1^{(t)}, \mathcal{V}_2^{(t)}, \mathcal{V}_3^{(t)}, \mathcal{V}_4^{(t-1)}) \\
&\overset{\text{(d)}}{\geq} \mathcal{F}(\mathcal{V}_1^{(t)}, \mathcal{V}_2^{(t)}, \mathcal{V}_3^{(t)}, \mathcal{V}_4^{(t)}),
\end{aligned} \tag{59}$$

Algorithm 3 Communication and Computation Design Algorithm for Federated Learning

1: Let $\mathcal{F}$ denote the objective function in problem (26).
2: Let $\mathcal{V}_1^{(t)}, \mathcal{V}_2^{(t)}, \mathcal{V}_3^{(t)}, \mathcal{V}_4^{(t)}$ denote the four groups of optimization variables in Sub-problem I, Sub-problem II, Sub-problem III, Sub-problem IV respectively. Here, t is the iteration number.
3: Initialize $\mathcal{V}_2^{(0)}, \mathcal{V}_3^{(0)}$ and $\mathcal{V}_4^{(0)}$
4: **repeat**
5: Given $\mathcal{V}_2^{(t-1)}, \mathcal{V}_3^{(t-1)}$ and $\mathcal{V}_4^{(t-1)}$, optimize $\mathcal{V}_1$ by solving sub-problem I according to Section V-A, and obtain the solution as $\mathcal{V}_1^{(t)}$;
6: Given $\mathcal{V}_1^{(t)}, \mathcal{V}_3^{(t-1)}$ and $\mathcal{V}_4^{(t-1)}$, optimize $\mathcal{V}_2$ by solving sub-problem II according to Section V-B, and obtain the solution as $\mathcal{V}_2^{(t)}$;
7: Given $\mathcal{V}_1^{(t)}, \mathcal{V}_2^{(t)}$ and $\mathcal{V}_4^{(t-1)}$ optimize $\mathcal{V}_3$ by solving sub-problem III according to Section V-C, and obtain the solution as $\mathcal{V}_3^{(t)}$;
8: Given $\mathcal{V}_1^{(t)}, \mathcal{V}_2^{(t)}$ and $\mathcal{V}_3^{(t)}$ optimize $\mathcal{V}_4$ by solving sub-problem IV according to Section V-D, and obtain the solution as $\mathcal{V}_4^{(t)}$;
9: **until** the improvement is negligible; i.e., $\mathcal{F}\left(\mathcal{V}_1^{(t)}, \mathcal{V}_2^{(t)}, \mathcal{V}_3^{(t)}, \mathcal{V}_4^{(t)}\right) \leq \epsilon_0$ for a given small value ϵ_0, or the number of iterations achieves some given T.

where steps (a), (b), (c) and (d) above follow from Lines 5, 6, 7 and 8 of Algorithm 3, respectively. In addition, $\mathcal{F}(\mathcal{V}_1, \mathcal{V}_2, \mathcal{V}_3, \mathcal{V}_4)$ in each iteration; i.e., $\mathcal{F}(\mathcal{V}_1^{(t)}, \mathcal{V}_2^{(t)}, \mathcal{V}_3^{(t)}, \mathcal{V}_4^{(t)})$ clearly has a lower bound of 0. From the monotone convergence theorem, any non-increasing and lower-bounded sequence converges. Hence, the sequence $\mathcal{F}(\mathcal{V}_1^{(t)}, \mathcal{V}_2^{(t)}, \mathcal{V}_3^{(t)}, \mathcal{V}_4^{(t)})|_{t=1,2,3\ldots}$ converges. This means our Algorithm 3 converges.

Computational complexity. For sub-problem I, sub-problem II and sub-problem III, we use the subgradient projection method to obtain the Lagrange multipliers in each step. Let ϵ_1 be the achieved accuracy, and i_1 be the maximum iterations in the subgradients projection methods, and then the computational complexity is $\mathcal{O}\left(i_1 \frac{1}{\epsilon_1^2}\right)$.

For sub-problem IV, if we utilize some convex solver based on interior-point methods such as CVX [43], then the worst-case complexity for each CCCP iteration can be given by $\mathcal{O}(N+M)^{3.5}$, where N and M are the number of UEs and the number of edge servers. Let i_2 be the number of CCCP iterations, and the computational complexity can be $\mathcal{O}(i_2(N+M)^{3.5})$.

With the aforementioned computational complexity for sub-problem I to sub-problem IV, let i_3 be the number of total iterations in the BCD algorithm. Then, the overall computational complexity is $\mathcal{O}\left(i_3\left(i_1\frac{1}{\epsilon_1^2} + i_2(N+M)^{3.5}\right)\right)$. The complexity is dominated by the subgradient projection and convex optimization steps, which scale polynomially with the number of UEs and edge servers. The computational overhead of the proposed optimization process is minimal compared to the computational cost of FL training. FL inherently involves multiple rounds of training and communication across devices, which are computationally intensive processes. Therefore, the optimization overhead is expected to contribute only a small fraction of the total computational cost while yielding significant energy savings and improved training efficiency. Nonetheless, for scenarios requiring even lower overhead, lightweight heuristic-based optimization methods could be explored in future work to further reduce computational demands while maintaining resource efficiency.

VII. Numerical Results

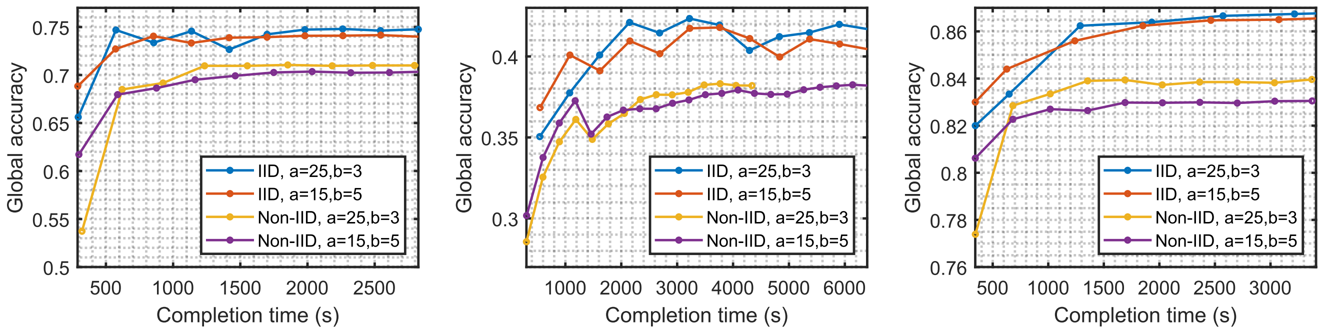
Now, we present numerical experiments to verify the performance of our solutions. The advantages of hierarchical federated learning system over edge-based federated learning system and cloud-based federated learning system have been investigated in [24]. In this paper, we evaluate the performance of the proposed algorithm on the hierarchical federated learning in terms of completion time and total energy consumption.

A. Experiment Settings

For simulations, we consider a hierarchical federated learning system with multiple user equipments, edge servers and one cloud server. The user equipments are deployed in a square area of size 500 m $\times$500 m with the edge servers located in the center and all the edge servers are deployed in an area with the cloud server located in the center. Similar to the approach in [51], we normalize constant γ in (15) to 1 when calculating the communication and computation expenditure. For simplicity, we use the free-space path loss model in [52]. Then with `wavelength` being the wavelength of the wireless signal and `distance` being the distance between UE n and edge server m, it holds that $g_{n,m} = \left(\frac{\texttt{wavelength}}{4\pi \times \texttt{distance}}\right)^2$. We set the frequency at 2 GHz so that $\texttt{wavelength} = \frac{3\times 10^8}{2\times 10^9} = \frac{3}{20}$m. Then $g_{n,m} = \left(\frac{3/20}{4\pi \times \texttt{distance}}\right)^2$. The number of CPU cycles required for UE n to compute one sample data C_n is set to be uniformly distributed in $[1,3] \times 10^4$. The effective capacitance coefficient of UE is $\kappa = 10^{-28}$. Although the algorithm is designed for the independent and identical (IID) data distribution case, we also conduct experiments on non-IID data distribution case. We use the Dirichlet distribution [53] as a measure of the degree of non-IIDness with the concentration parameter `alpha` being 1. For the parameters in the optimization problem regarding Assumption 1, L-smooth and β-strongly convex, we follow the experiment setting in [13]. The static assumptions in the proposed system model provide a theoretical foundation for understanding key trade-offs between energy consumption and latency. It enables the derivation of optimal resource allocation strategies under controlled conditions. While the framework assumes fixed parameters, it can be periodically reinitialized to adapt to gradual changes in dynamic environments. Additionally, integrating the framework with real-time network monitoring tools in practical deployments could allow dynamic adjustments to resource allocation in real-world scenarios.

B. The Optimal Values of Local Computations and Edge Aggregations

Firstly, we train the VGG-16 network on three different datasets with different configurations of a and b on both



(a) CIFAR-10 dataset (b) CIFAR-100 dataset (c) Fashion-MNIST dataset

Fig. 2. Comparison of global accuracy vs. completion time for different configurations of a and b across multiple datasets with IID and non-IID data distribution.

IID and non-IID data distribution cases. The experiments are conducted with 4 edge servers and 16 UEs, providing a realistic hierarchical federated learning setup. Fig. 2 illustrates the impact of various configurations of local and edge iterations on global accuracy and training completion time using CIFAR-10, CIFAR-100 and Fashion-MNIST datasets, where each subfigure corresponds to one dataset. It can be observed that for each dataset, the lines display varying slopes and intersections, indicating that different configurations result in different trade-offs between accuracy and time for both IID and non-IID cases. Some configurations may achieve higher accuracy quickly, while others might take longer to converge to the same level of accuracy. For instance, for the CIFAR-10 dataset in Fig. 2 (a), when the target global accuracy is less than 0.7 for the IID case and less than 0.68 for the non-IID case, the configuration "$a = 15$, $b = 5$" will achieve the target global accuracy with minimal time. When the target global accuracy is larger than 0.71 for the IID case, and larger than 0.68 for the non-IID case, "$a = 25$, $b = 3$" will be the time-optimal configuration. This variability underscores that achieving a desired global accuracy in the minimum time requires careful optimization of the local and edge iterations.

Next, from Fig. 3 to Fig. 9, we use the CIFAR-10 dataset with a VGG-16 network under an IID data distribution to evaluate the proposed optimization algorithm, aligning with the conditions for which it was designed. In this simulation, we deploy 5 edge servers, each associated with 20 UEs. The maximum transmission power p_n^{max} is 10 dBm for each device. To achieve a given global accuracy ϵ within minimum time, we optimize the local iterations and edge iterations. Fig. 3 and Fig. 4 show the completion time and energy consumption under different target global accuracy ϵ. The CPU frequency, transmission power, bandwidth and UE-to-edge association are all optimized here. Both weight parameters ρ_1 and ρ_2 are set to be 0.5. The proposed algorithm is denoted as "Optimal a and b". We compare the proposed algorithm with three other schemes

- **Optimal a only:** The "Optimal a only" method only optimizes local computation a while choosing random edge iteration b.
- **Optimal b only:** The "Optimal b only" method only optimizes edge iteration b while performing random local computation b.
- **Baseline:** The baseline method picks random local computation a and edge iteration b.

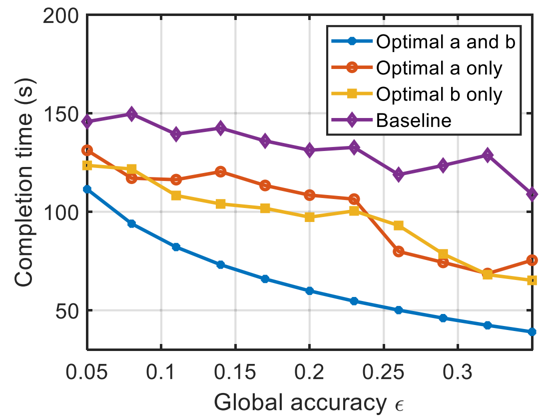


Fig. 3. Completion time under different target global accuracy.

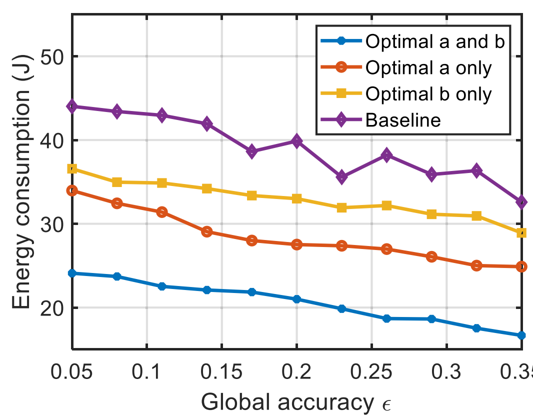


Fig. 4. Energy consumption under different target global accuracy.

Fig. 3 shows that when global accuracy ϵ increases (i.e., lower machine learning model accuracy is required), all four strategies complete the FL process within less time. The proposed method (i.e., "Optimal a and b") achieves the shortest time, while the baseline uses the most time to achieve a given global accuracy. In Fig. 4, the energy consumption is displayed under different global accuracy ϵ requirements. The proposed algorithm always achieves the lowest energy consumption, while the baseline scheme always consumes the most energy. The "Optimal a only" strategy performs a little better than the "Optimal b only" strategy in terms of energy consumption. It is

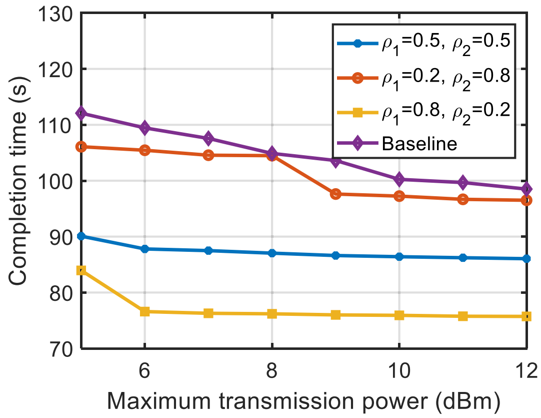


Fig. 5. Completion time with different weight parameters under different maximum available transmission power.

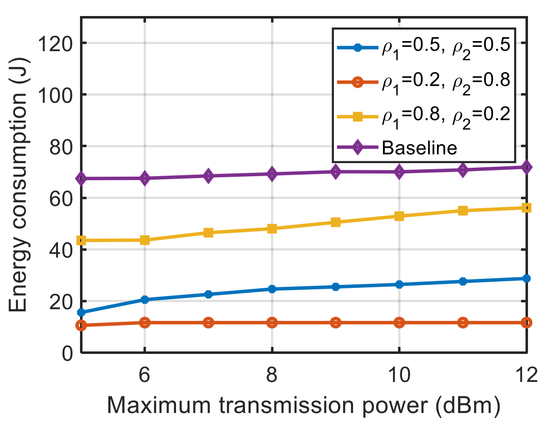


Fig. 6. Energy consumption with different weight parameters under different maximum available transmission power.

because the local computation energy covers a large proportion of total energy consumption. When the local computation a is optimized, the energy consumption for local computation is reduced.

C. The Trade-off Between FL Completion Time and Energy Consumption

In Fig. 5 and Fig. 6, we compare the FL completion time and total energy consumption under different maximum available transmission power. In the simulation, we compare three pairs of $(\rho_1, \rho_2) = (0.5, 0.5)$, $(0.2, 0.8)$ and $(0.8, 0.2)$. The "Baseline" chooses random transmission power within the maximum available transmission power. The target global accuracy ϵ is 0.1. The observation is that when ρ_1 increases and ρ_2 decreases, the FL completion time becomes shorter and the energy becomes higher, and vice versa. That's because when ρ_1 gets smaller, the algorithm emphasizes more on minimizing the FL completion time. Besides, in Fig. 5, the red line is below the purple line. It means even when ρ_1 is set to be very small, e.g., $\rho_1 = 0.2$, our proposed algorithm still outperforms the baseline in terms of FL completion time. Similarly, in Fig. 6, the yellow line is located below the purple line. It indicates that even if we set ρ_2 to be 0.2 (i.e., the system pays little attention to energy consumption), our proposed strategy continues to perform better than the baseline. This simulation demonstrates that our framework can adapt to varying operational constraints and priorities by dynamically adjusting the values of ρ_1 and ρ_2. When time minimization is prioritized, a larger ρ_1 relative to ρ_2 can be selected to ensure that the optimization heavily favors reducing time, which is particularly beneficial for latency-sensitive applications. Conversely, when energy efficiency is more critical, such as in battery-constrained IoT devices, increasing ρ_2 relative to ρ_1 shifts the focus towards minimizing energy consumption.

Fig. 7 provides a comparison between the proposed method and existing approaches that focus exclusively on optimizing either energy consumption or delay. In the figure, the scatter points are color-coded to represent varying levels of maximum transmission power, as indicated by the color bar on the right. The proposed method in this figure is conducted with $\rho_1 = \rho_2 = 0.5$. The results clearly demonstrate that the proposed method achieves a balanced trade-off between energy and delay, as shown in Fig. 7 (a). The energy-only optimization method achieves minimal energy consumption but at the cost of significantly higher delays. Conversely, the delay-only optimization method minimizes delay but results in much higher energy consumption. Fig. 7 (b), (c), and (d) provide zoomed-in views of these three lines in specific operational regions. These findings demonstrate that the proposed method can effectively balance both energy efficiency and delay minimization in diverse scenarios.

D. The Optimal UE-to-Edge Association

In this section, we evaluate three different UE-to-edge association strategies under the global accuracy requirement $\epsilon = 0.25$. The strategies under examination include the proposed method, the greedy algorithm, and the random UE-to-edge association strategy:

- **Proposed method:** Our proposed method leverages CCCP technique through solving an optimization framework that takes into account both the bandwidth constraints and the accuracy requirements. It aims to maximize the overall system performance by efficiently allocating UEs to edge servers, ensuring an optimal balance between resource utilization and accuracy.
- **Greedy algorithm:** The greedy algorithm adopts a straightforward approach where UEs are selected based on the maximum transmission rate under the given bandwidth constraints for each edge server. Specifically, it iteratively assigns the UEs with the highest transmission rates to the available edge servers until the bandwidth limit is reached. This method prioritizes high transmission rates, potentially leading to suboptimal overall performance due to its myopic nature.
- **Random UE-to-edge association:** The random UE-to-edge association assigns UEs to all the edge servers randomly, adhering only to the bandwidth constraints.

Fig. 8 illustrates the system's maximum latency under different numbers of edge servers. The number of total UEs is 100. It can be found that the proposed method consistently achieves the lowest latency under different numbers of edge servers. The greedy algorithm always chooses the UEs with the maximum transmission rate available, thus outperforming

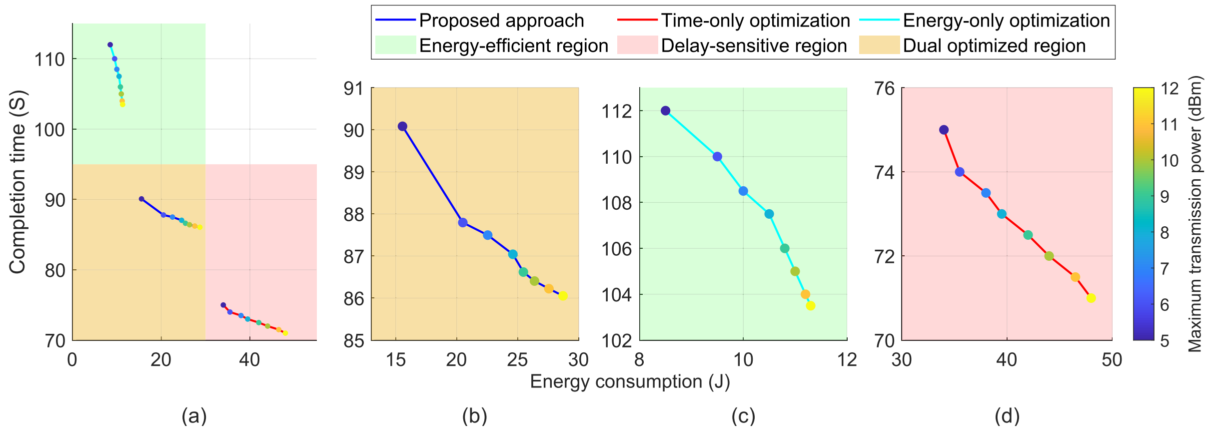


Fig. 7. (a) Overview of energy consumption vs. completion time across different transmission powers. (b) Zoomed-in view: trade-offs in the dual-optimized region. (c) Zoomed-in view: energy-efficient region. (d) Zoomed-in view: delay-sensitive region.

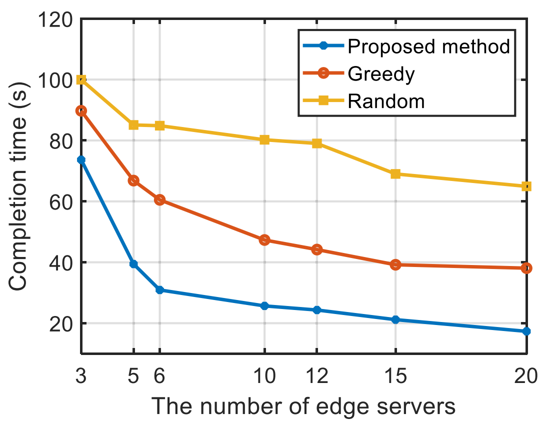


Fig. 8. Completion time under different numbers of edge servers.

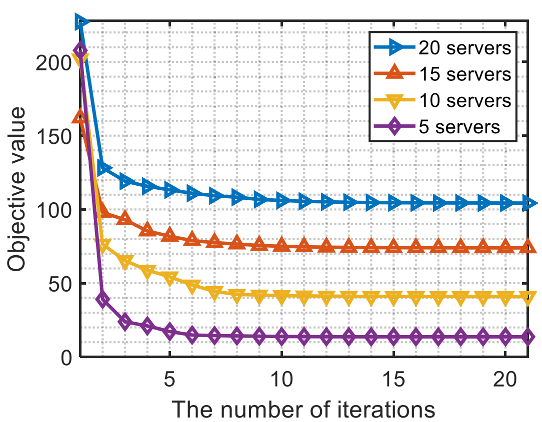


Fig. 9. Convergence performance of the proposed algorithm.

the random association. It is noticed that when the number of edge servers is smaller, the latency is higher. That is because when the number of edge servers is small, UEs have few choices. More UEs have to associate with edge servers whose transmission rate is large.

In Fig 9, we assess the convergence performance of the algorithm for the UE-to-edge association problem. The numbers of edge servers are set to 20, 15, 10 and 5, while other parameters are kept default. Once the objective value stabilizes, it indicates that the algorithm has reached a stationary point. It can be found that scenarios with fewer servers demonstrate faster convergence. One primary reason for this acceleration is the direct correlation between the number of servers and the complexity of the associated optimization problem. Specifically, fewer servers mean there are fewer constraints to consider and fewer parameters to adjust during the optimization process, facilitating faster solutions. In each scenario, our algorithm converged within 10 iterations, which indicates that our algorithm can effectively solve problem (55).

VIII. Conclusion

In this paper, we have investigated the problem of computation and communication resource optimization in a hierarchical federated learning framework. In particular, we have formulated a joint computation and communication optimization problem aimed at minimizing the weighted sum of energy consumption and FL completion time, while still maintaining a predetermined level of accuracy. To tackle the complexity of this problem, we have methodically decomposed it into four distinct sub-problems, each addressing different aspects of the optimization challenge. Our approach employs sophisticated fractional programming techniques and the CCCP to effectively resolve these sub-problems, leveraging the strengths of each technique but also ensuring that the solutions are both practical and optimal within the constraints of the system. Simulation results underscore the efficiency of our proposed solution in achieving a balanced trade-off between FL completion time and total energy consumption. However, several limitations should be acknowledged. The framework assumes idealized communication conditions, which may not account for unpredictable network disruptions or delays in real-world scenarios. Moreover, the algorithm currently does not consider the impact of adversarial attacks or security challenges that may arise in federated learning settings. Another notable challenge is the scalability of the proposed framework to large-scale FL systems. To address these challenges, future work could explore adaptive mechanisms to handle dynamic network conditions and incorporate security measures to mitigate adversarial risks. Furthermore, parallel processing techniques could be integrated into our optimization algorithm to enhance scalability. By distributing and executing optimization tasks concurrently at different hierarchical levels (e.g., UEs, edge servers, and cloud), computational overhead can be reduced while maintaining the framework's theoretical guarantees.

APPENDIX
PROOF OF THEOREM 1

With $F_n(\omega_n)$ being L-smooth in Assumption 1, according to the descent lemma, whose proof can be found in Proposition A.24 in [54], we have:

$$F_n(\omega_n^{(a+1)}) \leq F_n(\omega_n^{(a)}) + \nabla F_n(\omega_n^{(a)})^\intercal(\omega_n^{(a+1)} - \omega_n^{(a)}) + \frac{L}{2}\|\omega_n^{(a+1)} - \omega_n^{(a)}\|^2. \tag{A.1}$$

Since $\omega_n^{(a+1)} = \omega_n^{(a)} - \eta\nabla F_n(\omega_n^{(a)})$, (A.1) is equivalent to

$$\begin{aligned} &F_n(\omega_n^{(a+1)}) \\ &\leq F_n(\omega_n^{(a)}) - \eta\|\nabla F_n(\omega_n^{(a)})\|^2 + \frac{L\eta^2}{2}\|\nabla F_n(\omega_n^{(a)})\|^2 \\ &= F_n(\omega_n^{(a)}) - \eta\left(1 - \frac{L\eta}{2}\right)\|\nabla F_n(\omega_n^{(a)})\|^2. \end{aligned} \tag{A.2}$$

With $F_n(\omega_n)$ being β-strongly convex in Assumption 1, we have:

$$\begin{aligned} &F_n(\omega_n) \\ &\geq F_n(\omega_n^{(a)}) + \nabla F_n(\omega_n^{(a)})^\intercal(\omega_n - \omega_n^{(a)}) + \frac{\beta}{2}\|\omega_n - \omega_n^{(a)}\|^2. \end{aligned} \tag{A.3}$$

Letting $\omega_n = \omega_n^*$ as the minimizer for both sides yields:

$$\begin{aligned} F_n(\omega_n^*) &\geq F_n(\omega_n^{(a)}) - \frac{\nabla^2 F_n(\omega_n^{(a)})}{\beta} + \frac{\nabla^2 F_n(\omega_n^{(a)})}{2\beta} \\ &= F_n(\omega_n^{(a)}) - \frac{\|\nabla F_n(\omega_n^{(a)})\|^2}{2\beta}. \end{aligned} \tag{A.4}$$

Thus, we have

$$\|\nabla F_n(\omega_n^{(a)})\|^2 \geq 2\beta[F_n(\omega_n^{(a)}) - F_n(\omega_n^*)], \tag{A.5}$$

where ω_n^* is the optimal solution to problem (1). Based on (A.2) and (A.5), it can be proved that:

$$\begin{aligned} &F_n(\omega_n^{(a+1)}) - F_n(\omega_n^*) \\ &\overset{(A.2)}{\leq} F_n(\omega_n^{(a)}) - \eta\left(1 - \frac{L\eta}{2}\right)\|\nabla F_n(\omega_n^{(a)})\|^2 - F_n(\omega_n^*) \\ &\overset{(A.5)}{\leq} F_n(\omega_n^{(a)}) - 2\beta\eta\left(1 - \frac{L\eta}{2}\right)[F_n(\omega_n^{(a)}) - F_n(\omega_n^*)] - F_n(\omega_n^*) \\ &= \left[1 - 2\beta\eta\left(1 - \frac{L\eta}{2}\right)\right][F_n(\omega_n^{(a)}) - F_n(\omega_n^*)] \\ &\leq \left[1 - 2\beta\eta\left(1 - \frac{L\eta}{2}\right)\right]^{a+1}[F_n(\omega_n^{(0)}) - F_n(\omega_n^*)]. \end{aligned} \tag{A.6}$$

Considering that $\forall x \geq 0, e^{-x} \geq 1 - x$ and (A.6), we can get:

$$\begin{aligned} &F_n(\omega_n^{(a+1)}) - F_n(\omega_n^*) \leq \\ &e^{-(a+1)2\beta\eta\left(1 - \frac{L\eta}{2}\right)} \cdot [F_n(\omega_n^{(0)}) - F_n(\omega_n^*)]. \end{aligned} \tag{A.7}$$

Therefore, in order to achieve a local accuracy θ, (7) holds.

ACKNOWLEDGMENT

Improvements over [1] are explicitly discussed in this article. Any opinions, findings and conclusions or recommendations expressed in this material are those of the author(s) and do not reflect the views of National Research Foundation, Singapore.

REFERENCES

[1] C. Liu, T. J. Chua, and J. Zhao, "Time minimization in hierarchical federated learning," in *Proc. IEEE/ACM 7th Symp. Edge Comput. (SEC)*, Dec. 2022, pp. 96–106.

[2] K. Bonawitz et al., "Towards federated learning at scale: System design," *Proc. Mach. Learn. Syst.*, pp. 374–388, Aug. 2019.

[3] F. Haddadpour and M. Mahdavi, "On the convergence of local descent methods in federated learning," 2019, *arXiv:1910.14425*.

[4] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated learning with non-IID data," 2018, *arXiv:1806.00582*.

[5] X. Cao et al., "Communication-efficient distributed learning: An overview," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 4, pp. 851–873, Apr. 2023.

[6] Y. Qu et al., "Decentralized federated learning for UAV networks: Architecture, challenges, and opportunities," *IEEE Netw.*, vol. 35, no. 6, pp. 156–162, Nov. 2021.

[7] M. Chen, H. V. Poor, W. Saad, and S. Cui, "Convergence time optimization for federated learning over wireless networks," *IEEE Trans. Wireless Commun.*, vol. 20, no. 4, pp. 2457–2471, Apr. 2021.

[8] W. Shi, S. Zhou, and Z. Niu, "Device scheduling with fast convergence for wireless federated learning," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2020, pp. 1–6.

[9] M. Chen, Z. Yang, W. Saad, C. Yin, H. V. Poor, and S. Cui, "A joint learning and communications framework for federated learning over wireless networks," *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 269–283, Jan. 2021.

[10] S. Wang et al., "Adaptive federated learning in resource constrained edge computing systems," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 6, pp. 1205–1221, Jun. 2019.

[11] M. Mohammadi Amiri and D. Gündüz, "Machine learning at the wireless edge: Distributed stochastic gradient descent over-the-air," *IEEE Trans. Signal Process.*, vol. 68, pp. 2155–2169, 2020.

[12] M. Chen, U. Challita, W. Saad, C. Yin, and M. Debbah, "Artificial neural networks-based machine learning for wireless networks: A tutorial," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3039–3071, 4th Quart., 2019.

[13] N. H. Tran, W. Bao, A. Zomaya, M. N. H. Nguyen, and C. S. Hong, "Federated learning over wireless networks: Optimization model design and analysis," in *Proc. IEEE Conf. Comput. Commun.*, Apr. 2019, pp. 1387–1395.

[14] H. Wang, S. Sievert, Z. Charles, S. Liu, S. Wright, and D. Papailiopoulos, "ATOMO: Communication-efficient learning via atomic sparsification," 2018, *arXiv:1806.04090*.

[15] J. Konečny, H. B. McMahan, F. X. Yu, P. Richtárik, A. T. Suresh, and D. Bacon, "Federated learning: Strategies for improving communication efficiency," 2016, *arXiv:1610.05492*.

[16] A. F. Aji and K. Heafield, "Sparse communication for distributed gradient descent," 2017, *arXiv:1704.05021*.

[17] Z. Yang, M. Chen, W. Saad, C. S. Hong, and M. Shikh-Bahaei, "Energy efficient federated learning over wireless communication networks," *IEEE Trans. Wireless Commun.*, vol. 20, no. 3, pp. 1935–1949, Mar. 2021.

[18] Q. Zeng, Y. Du, K. Huang, and K. K. Leung, "Energy-efficient radio resource allocation for federated edge learning," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, Jun. 2020, pp. 1–6.

[19] Z. Yang et al., "Delay minimization for federated learning over wireless communication networks," in *Proc. Int. Conf. Mach. Learn.*, 2020.

[20] S. Luo, X. Chen, Q. Wu, Z. Zhou, and S. Yu, "HFEL: Joint edge association and resource allocation for cost-efficient hierarchical federated edge learning," *IEEE Trans. Wireless Commun.*, vol. 19, no. 10, pp. 6535–6548, Oct. 2020.

[21] Y. Cui, K. Cao, J. Zhou, and T. Wei, "Optimizing training efficiency and cost of hierarchical federated learning in heterogeneous mobile-edge cloud computing," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 42, no. 5, pp. 1518–1531, May 2023.

[22] W. Y. B. Lim, J. S. Ng, Z. Xiong, D. Niyato, C. Miao, and D. I. Kim, "Dynamic edge association and resource allocation in self-organizing hierarchical federated learning networks," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 12, pp. 3640–3653, Dec. 2021.

[23] L. Liu, J. Zhang, S. Song, and K. B. Letaief, "Hierarchical federated learning with quantization: Convergence analysis and system design," *IEEE Trans. Wireless Commun.*, vol. 22, no. 1, pp. 2–18, Jan. 2023.

[24] L. Liu, J. Zhang, S. H. Song, and K. B. Letaief, "Client-edge-cloud hierarchical federated learning," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2020, pp. 1–6.

[25] J. Wang, S. Wang, R. Chen, and M. Ji, "Demystifying why local aggregation helps: Convergence analysis of hierarchical SGD," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 8, pp. 8548–8556.

[26] N. Mhaisen, A. A. Abdellatif, A. Mohamed, A. Erbad, and M. Guizani, "Optimal user-edge assignment in hierarchical federated learning based on statistical properties and network topology constraints," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 1, pp. 55–66, Jan. 2022.

[27] A. A. Abdellatif et al., "Communication-efficient hierarchical federated learning for IoT heterogeneous systems with imbalanced data," *Future Gener. Comput. Syst.*, vol. 128, pp. 406–419, Mar. 2022.

[28] X. Huang, Y. Wu, C. Liang, Q. Chen, and J. Zhang, "Distance-aware hierarchical federated learning in blockchain-enabled edge computing network," *IEEE Internet Things J.*, vol. 10, no. 21, pp. 19163–19176, Nov. 2023.

[29] Y. Wang, Y. Xu, Q. Shi, and T.-H. Chang, "Quantized federated learning under transmission delay and outage constraints," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 1, pp. 323–341, Jan. 2022.

[30] P. K. Sangdeh, C. Li, H. Pirayesh, S. Zhang, H. Zeng, and Y. T. Hou, "CF4FL: A communication framework for federated learning in transportation systems," *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 3821–3836, Jun. 2023.

[31] M. Chen, N. Shlezinger, H. V. Poor, Y. C. Eldar, and S. Cui, "Communication-efficient federated learning," *Proc. Nat. Acad. Sci. USA*, vol. 118, no. 17, 2021, Art. no. 2024789118.

[32] Y. Jin, L. Jiao, Z. Qian, S. Zhang, and S. Lu, "Budget-aware online control of edge federated learning on streaming data with stochastic inputs," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 12, pp. 3704–3722, Dec. 2021.

[33] X. Zhou, J. Zhao, H. Han, and C. Guet, "Joint optimization of energy consumption and completion time in federated learning," in *Proc. IEEE 42nd Int. Conf. Distrib. Comput. Syst. (ICDCS)*, Jul. 2022, pp. 1005–1017.

[34] T. Wu et al., "Joint edge aggregation and association for cost-efficient multi-cell federated learning," in *Proc. IEEE Conf. Comput. Commun.*, May 2023, pp. 1–10.

[35] R. Jin, X. He, and H. Dai, "Communication efficient federated learning with energy awareness over wireless networks," *IEEE Trans. Wireless Commun.*, vol. 21, no. 7, pp. 5204–5219, Jul. 2022.

[36] X. Ji, J. Tian, H. Zhang, D. Wu, and T. Li, "Joint device selection and bandwidth allocation for cost-efficient federated learning in industrial Internet of Things," *IEEE Internet Things J.*, vol. 10, no. 10, pp. 9148–9160, May 2023.

[37] Y. Jin et al., "Scheduling in-band network telemetry with convergence-preserving federated learning," *IEEE/ACM Trans. Netw.*, vol. 31, no. 5, pp. 2313–2328, Oct. 2023.

[38] F. Wang, L. Jiao, K. Zhu, X. Lin, and L. Li, "Toward sustainable AI: Federated learning demand response in cloud-edge systems via auctions," in *Proc. IEEE Conf. Comput. Commun.*, May 2023, pp. 1–10.

[39] B. Luo, X. Li, S. Wang, J. Huang, and L. Tassiulas, "Cost-effective federated learning in mobile edge networks," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 12, pp. 3606–3621, Dec. 2021.

[40] T. Zhao, F. Li, and L. He, "DRL-based joint resource allocation and device orchestration for hierarchical federated learning in NOMA-enabled industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 19, no. 6, pp. 7468–7479, Jun. 2023.

[41] Y. Mao, J. Zhang, and K. B. Letaief, "Dynamic computation offloading for mobile-edge computing with energy harvesting devices," *IEEE J. Sel. Areas Commun.*, vol. 34, no. 12, pp. 3590–3605, Dec. 2016.

[42] C. Ma et al., "Distributed optimization with arbitrary local solvers," *Optim. Methods Softw.*, vol. 32, no. 4, pp. 813–848, Jul. 2017.

[43] S. P. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge Univ. Press, 2004.

[44] P. Tseng, "Convergence of a block coordinate descent method for nondifferentiable minimization," *J. Optim. Theory Appl.*, vol. 109, no. 3, pp. 475–494, Jun. 2001.

[45] Y. Jong, "Practical global optimization algorithm for the sum-of-ratios problem," *Optim. Online*, pp. 1–21, Jul. 2012.

[46] S. Boyd, L. Xiao, and A. Mutapcic, "Subgradient methods," Stanford Univ., Stanford, CA, USA, Tech. Rep. EE392o, 2003, vol. 2004.

[47] A. Schrijver, *Theory of Linear and Integer Programming*. Hoboken, NJ, USA: Wiley, 1998.

[48] J. Zhao, L. Qian, and W. Yu, "Human-centric resource allocation in the metaverse over wireless communications," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 3, pp. 514–537, Mar. 2024.

[49] M. F. Hanif, Z. Ding, T. Ratnarajah, and G. K. Karagiannidis, "A minorization-maximization method for optimizing sum rate in the downlink of non-orthogonal multiple access systems," *IEEE Trans. Signal Process.*, vol. 64, no. 1, pp. 76–88, Jan. 2016.

[50] H. A. Le Thi, T. Pham Dinh, and H. V. Ngai, "Exact penalty and error bounds in DC programming," *J. Global Optim.*, vol. 52, no. 3, pp. 509–535, Mar. 2012.

[51] S. R. Pandey, N. H. Tran, M. Bennis, Y. K. Tun, A. Manzoor, and C. S. Hong, "A crowdsourcing framework for on-device federated learning," *IEEE Trans. Wireless Commun.*, vol. 19, no. 5, pp. 3241–3256, May 2020.

[52] A. Goldsmith, *Wireless Communications*. Cambridge, U.K.: Cambridge Univ. Press, 2005.

[53] T. Lin, L. Kong, S. U. Stich, and M. Jaggi, "Ensemble distillation for robust model fusion in federated learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 2351–2363.

[54] D. P. Bertsekas, *Nonlinear Programming*. Belmont, MA, USA: Athena Scientific, 1999.

Chang Liu (Member, IEEE) received the B.E. degree from the University of Electronic Science and Technology of China, Chengdu, China, in 2020, and the Ph.D. degree from Nanyang Technological University, Singapore, in 2025. His research interests include mobile edge computing, federated learning, and optimization.

Jun Zhao (Member, IEEE) received the bachelor's degree from Shanghai Jiao Tong University, China, in July 2010, and the Ph.D. degree in electrical and computer engineering from Carnegie Mellon University (CMU), USA, in May 2015, affiliating with CMU's renowned CyLab Security and Privacy Institute. He is currently an Assistant Professor with the College of Computing and Data Science (CCDS)—formerly known as the School of Computer Science and Engineering (SCSE), Nanyang Technological University (NTU), Singapore. Before joining NTU first as a Post-Doctoral Researcher and then as a Faculty Member, he was at Arizona State University as an Arizona Computing Post-Doctoral Best Practices Fellow. He was selected as the Best Editor for IEEE Wireless Communications Letters and IEEE Transactions on Cognitive Communications and Networking in 2023. He is an Editor of IEEE Transactions on Information Forensics and Security and other journals.

Yitong Wang received the bachelor's degree in communication engineering from Zhengzhou University, China, in 2021, and the master's degree in telecommunications from The Hong Kong University of Science and Technology (HKUST), Hong Kong, SAR, China, in 2022. He is currently pursuing the Ph.D. degree with the College of Computing and Data Science (CCDS), Nanyang Technological University, Singapore. His research interests include optimization, wireless communications, and physical layer security.